# MI DISP API

**Version 2.26**

# REVISION HISTORY

| Revision No. | Description | Date |
|---|---|---|
| 2.03 | • Initial release | 04/12/2018 |
| 2.04 | • Added the following APIs:<br>  • MI_DISP_GetLcdParam<br>  • MI_DISP_SetLcdParam<br>  • MI_DISP_DeviceGetColorTempeture<br>  • MI_DISP_DeviceSetColorTempeture<br>  • MI_DISP_DeviceSetGammaParam<br>  • MI_DISP_SetVideoLayerRotateMode<br>• Modified DISP data type:<br>  • MI_DISP_InputPortAttr_t<br>  with the following new parameters added:<br>  - MI_U16 u16SrcWidth;<br>  - MI_U16 u16SrcHeight; | 06/06/2019 |
| 2.05 | • Added the following APIs:<br>  • MI_DISP_DeviceAttach<br>  • MI_DISP_DeviceDetach | 06/19/2019 |
| 2.06 | • Modify MI_DISP_ClearInputPortBuffer parameter<br>• Added the following APIs:<br>  • MI_DISP_InitDev<br>  • MI_DISP_DeInitDev | 05/13/2020 |
| 2.07 | • Added new interface type | 06/18/2020 |
| 2.08 | • Modified chip's name<br>• Deleted the following APIs:<br>  • MI_DISP_IntfSync_e<br>  • MI_DISP_PixelFormat_e<br>  • MI_DISP_VideoFrame_t<br>• Added the following APIs:<br>  • MI_DISP_Interface_e<br>  • MI_DISP_OutputTiming_e<br>  • MI_DISP_SetZoomInWindow<br>• Modified some DISP data type | 08/04/2020 |
| 2.09 | • Added new interface type | 09/03/2020 |
| 2.10 | • Added new interface type<br>  • E_MI_DISP_INTF_SRGB | 10/26/2020 |
| 2.11 | • Added new interface type<br>  • MI_DISP_DevAttachAttr_t | 12/24/2020 |
| 2.12 | • Add APIs:<br>  • MI_DISP_SetWBCSource<br>  • MI_DISP_GetWBCSource | 01/05/2021 |

| Revision No. | Description | Date |
|---|---|---|
| | • MI_DISP_SetWBCAttr<br>• MI_DISP_GetWBCAttr<br>• MI_DISP_EnableWBC<br>• MI_DISP_DisableWBC | |
| 2.13 | • Added new interface type<br>• E_MI_DISP_INTF_MCU<br>• E_MI_DISP_INTF_MCU_NOFLM | 01/22/2021 |
| 2.14 | • Modified struct MI_DISP_SyncInfo_t<br>• Added new parameters<br>• MI_U16 u16VStart;<br>• MI_U16 u16HStart; | 04/16/2021 |
| 2.15 | • New timing type<br>• E_MI_DISP_OUTPUT_1920x1080_5994<br>• E_MI_DISP_OUTPUT_1920x1080_2997<br>• E_MI_DISP_OUTPUT_1280x720_5994 | 04/22/2021 |
| 2.16 | • New timing type<br>• E_MI_DISP_OUTPUT_1280x720_2997 | 05/17/2021 |
| 2.17 | • Added some info for Muffin | 05/25/2021 |
| 2.18 | • New timing type<br>• E_MI_DISP_OUTPUT_2560x1440_50<br>• E_MI_DISP_OUTPUT_2560x1440_60<br>• E_MI_DISP_OUTPUT_3840x2160_25 | 06/10/2021 |
| 2.19 | • Modified Ikayaki flow chart<br>• New timing type<br>• E_MI_DISP_OUTPUT_3840x2160_2997 | 08/05/2021 |
| | • Added PROCFS INTRODUCTION | 08/25/2021 |
| 2.20 | • Added some info for Mochi<br>• Error code optimization | 09/03/2021 |
| 2.21 | • Added support for BT601 output<br>• Added support for csc under CVBS interface | 10/12/2021 |
| 2.22 | • New timing type<br>• E_MI_DISP_OUTPUT_720P24<br>• E_MI_DISP_OUTPUT_720P25<br>• E_MI_DISP_OUTPUT_720P30<br>• E_MI_DISP_OUTPUT_1920x1080_2398<br>• E_MI_DISP_OUTPUt_3840x2160_24<br>• E_MI_DISP_OUTPUT_848x480_60<br>• E_MI_DISP_OUTPUT_1280x768_60<br>• E_MI_DISP_OUTPUT_1280x960_60<br>• E_MI_DISP_OUTPUT_1360x768_60 | 11/29/2021 |

| Revision No. | Description | Date |
|---|---|---|
| | • E_MI_DISP_OUTPUT_1400x1050_60<br>• E_MI_DISP_OUTPUT_1600x900_60<br>• E_MI_DISP_OUTPUT_1920x1440_60 | |
| 2.23 | • New interface type<br>  • E_MI_DISP_INTF_BT1120_DDR | 12/08/2021 |
| 2.24 | • New interface<br>  • MI_DISP_SetVideoLayerAttrBatchBegin<br>  • MI_DISP_SetVideoLayerAttrBatchEnd<br>• Modify the notes for the following interfaces<br>  • MI_DISP_DisableVideoLayer<br>  • MI_DISP_UnBindVideoLayer | 12/20/2021 |
| 2.25 | • Modify 1.2.6 Flow Chart<br>• Modify 3.23 Note | 2/16/2022 |
| 2.26 | • Add some info of Maruko | 3/7/2022 |

# TABLE OF CONTENTS

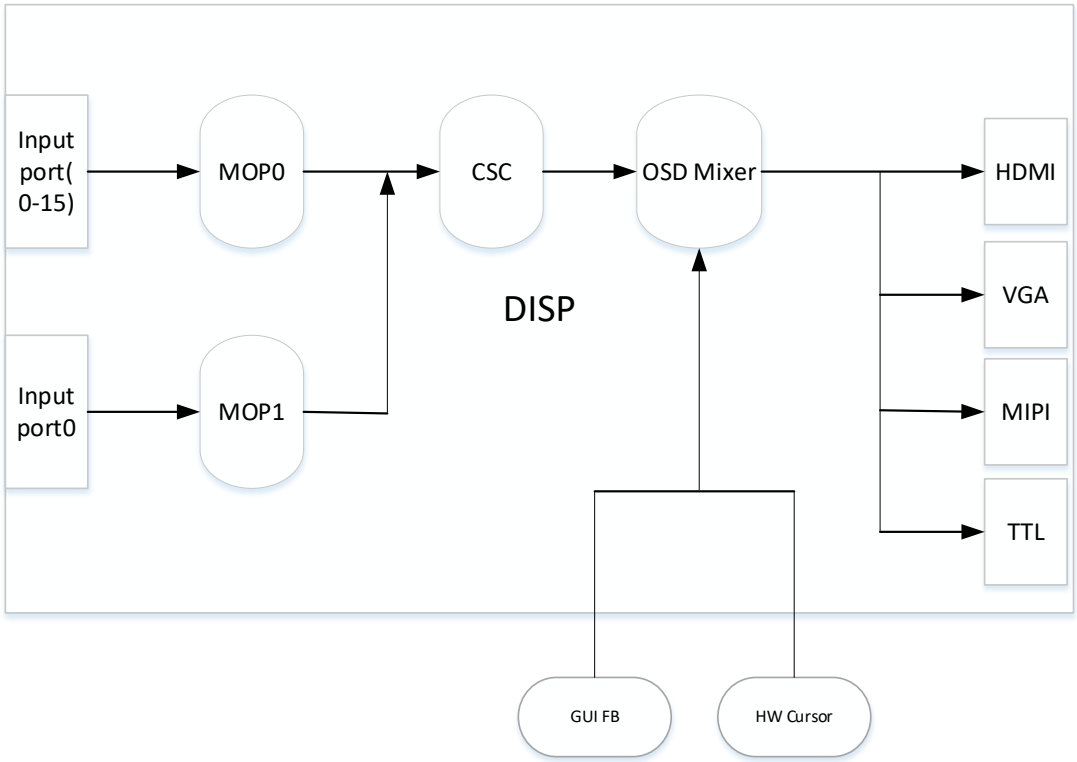
# 1. SUMMARY

## 1.1. Module Description

The DISP is a video display unit. Its main function is to splice the image output from modules which bind with it, and to perform color space conversion on the image. After that, the image is output to the display or LCD through HDMI/VGA/MIPI/TTL or other interfaces.

## 1.2. Flow Chart

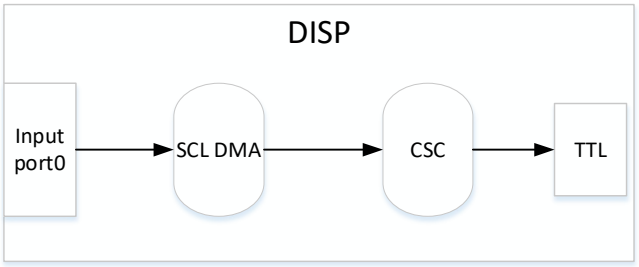
### 1.2.1. Taiyaki/Takoyaki



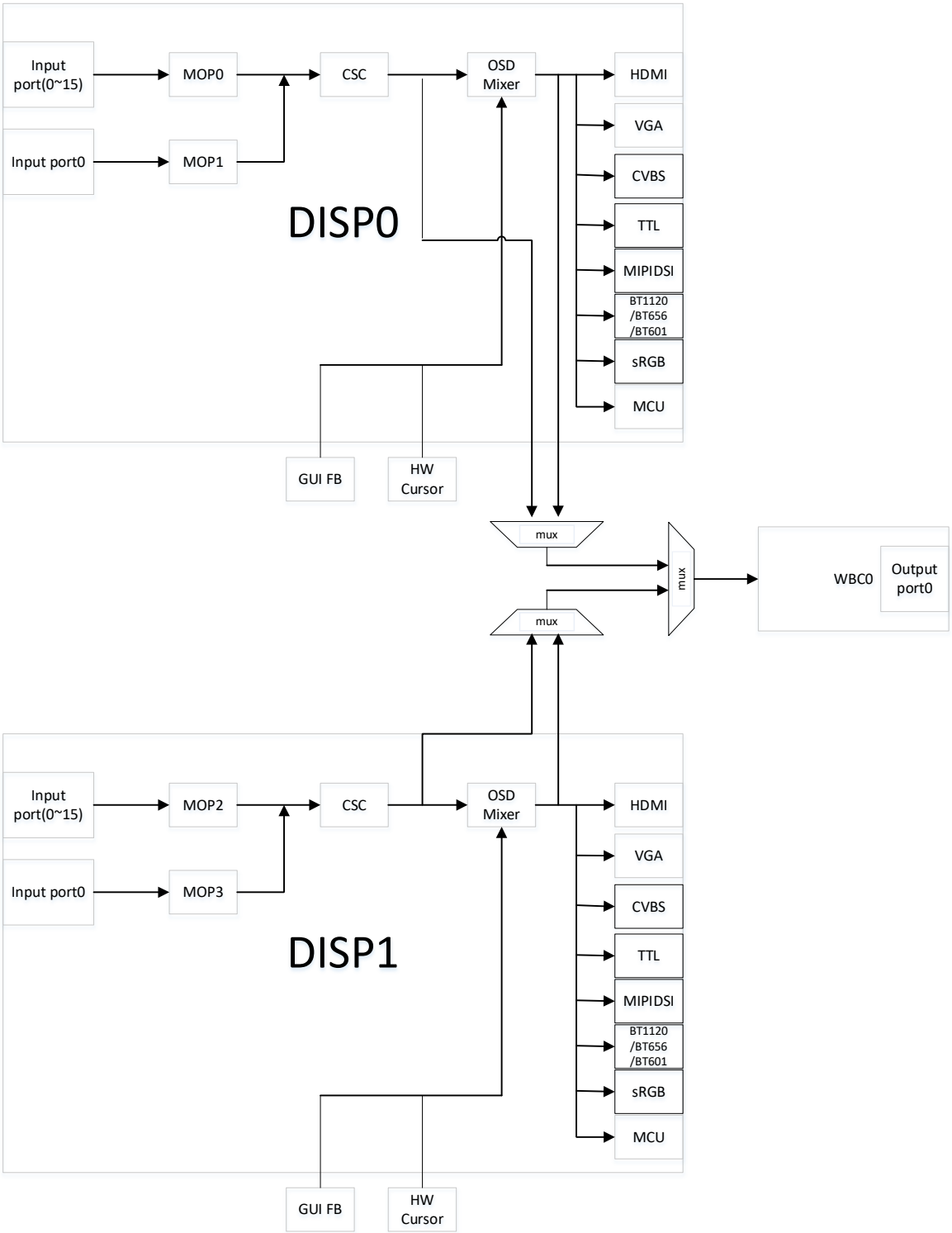

※ Note

1. HDMI and VGA interface can be used at the same time to output an image, whereas MIPI/TTL interface can only be used one at a time to output an image.
2. The output screen of MOP1 will be superimposed on MOP0 to realize the function of PIP, that is, the display priority of MOP1 is higher than that of MOP0.
3. The display positions of the 16 Input Ports on MOP0 cannot overlap each other.

### 1.2.2. Pretzel / Pudding



### 1.2.3. Tiramisu



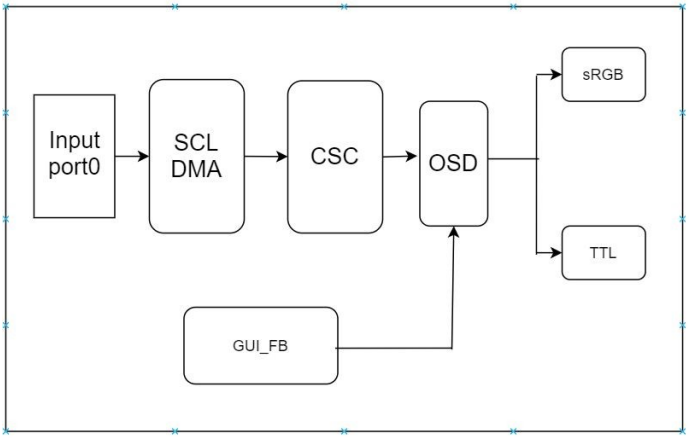

※ Note

1. For each device, HDMI and VGA interfaces can be used to output an image at the same time, whereas interfaces like CVBS, MIPI, TTL, BT1120, BT656, BT601, sRGB, MCU can only be output an image separately.
2. The output screen of MOP1 will be superimposed on MOP0 to realize the function of PIP, that is, the

display priority of MOP1 is higher than that of MOP0.

3. The display positions of the 16 Input Ports on MOP0 cannot overlap each other.
4. MOP2 and MOP3 are used in the same way as MOP0 and MOP1.
5. The output of DISP0 and DISP1 can be written back to the memory by WBC.

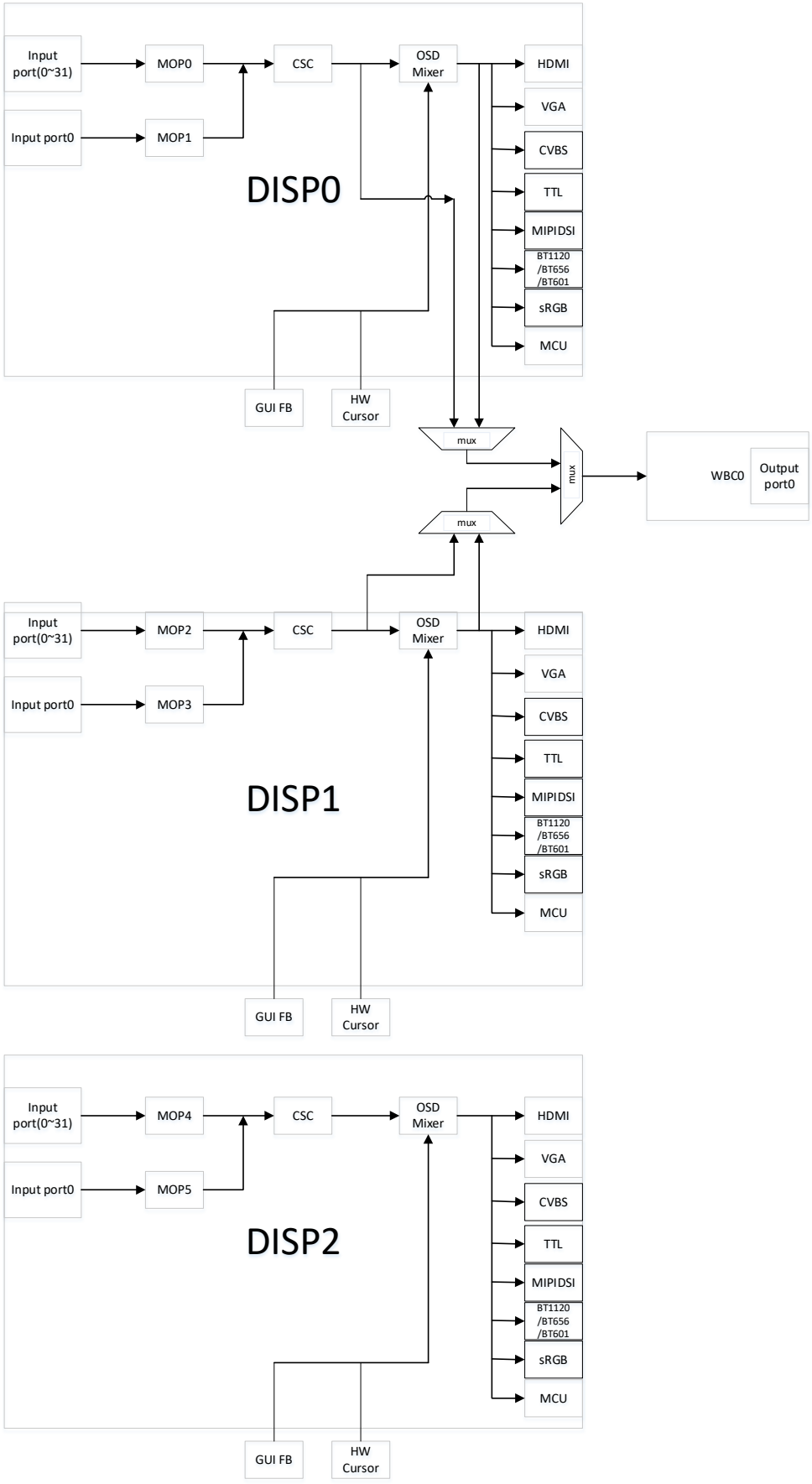
### 1.2.4. Ikayaki




※ Note

TTL/sRGB interface can only be used one to output an image at a time.

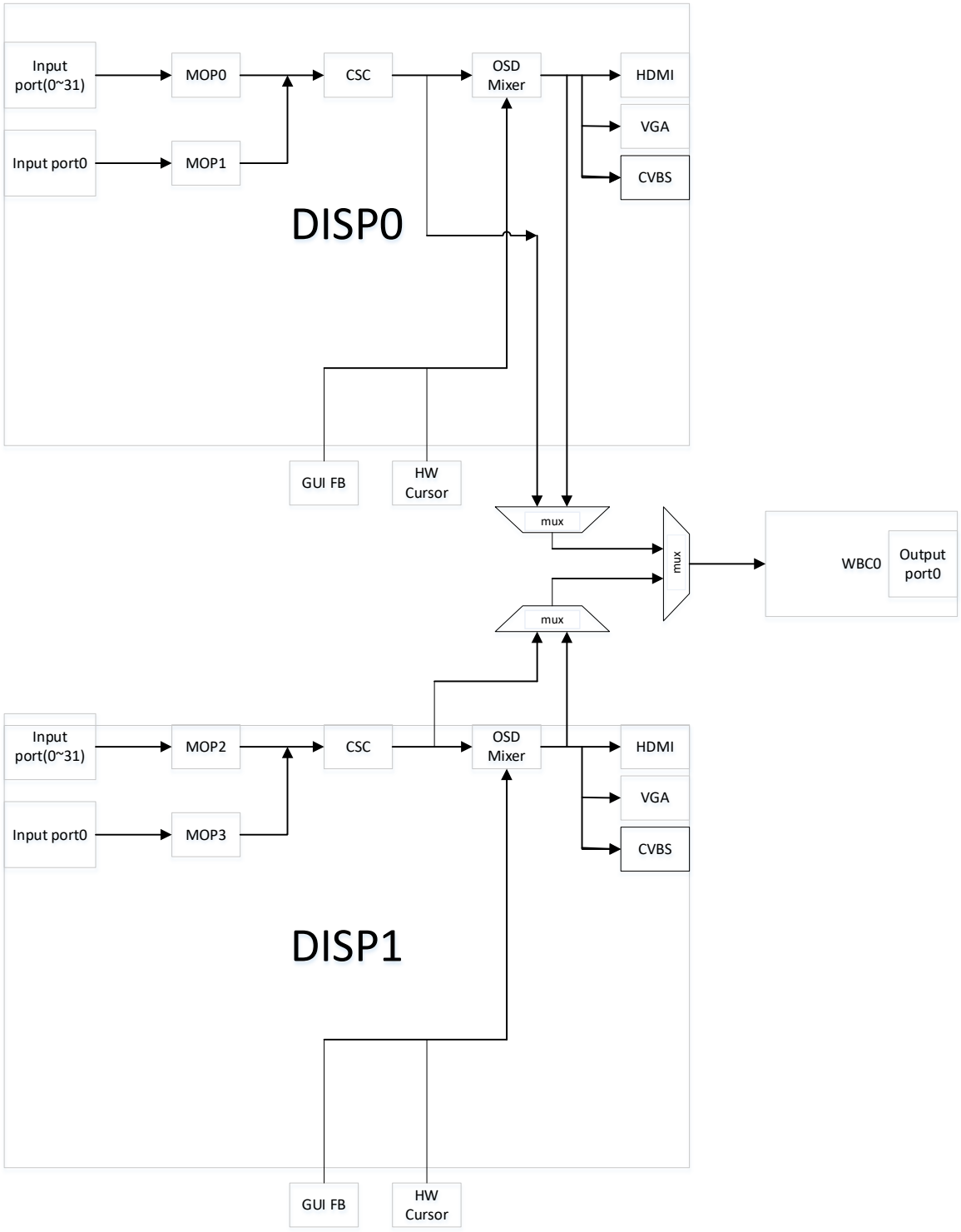
### 1.2.5. Muffin



※ Note

1. For each device, HDMI and VGA interfaces can be used to output an image at the same time, whereas interfaces like CVBS, MIPI, TTL, BT1120, BT656, BT601, sRGB, MCU can only be output an image separately.
2. The output screen of MOP1 will be superimposed on MOP0 to realize the function of PIP, that is, the display priority of MOP1 is higher than that of MOP0.
3. The display positions of the 32 Input Ports on MOP0 cannot overlap each other.
4. MOP2/3 and MOP4/5 are used in the same way as MOP0 and MOP1.
5. The output of DISP0 and DISP1 can be written back to the memory by WBC.

### 1.2.6. Mochi

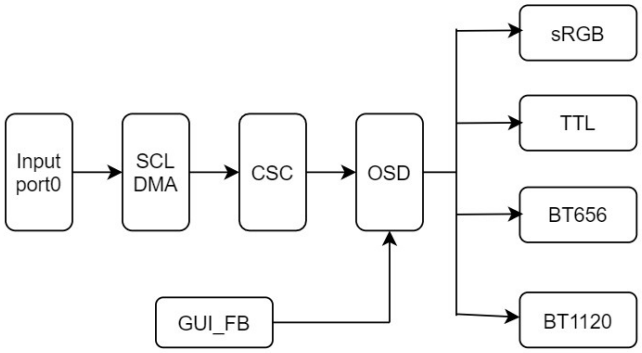



※ Note

1. For each device, HDMI and VGA interfaces can be used to output an image at the same time, whereas interfaces like CVBS can only be output an image separately.
2. The output screen of MOP1 will be superimposed on MOP0 to realize the function of PIP, that is, the display priority of MOP1 is higher than that of MOP0.

3. The display positions of the 32 Input Ports on MOP0 cannot overlap each other.
4. MOP2 and MOP3 are used in the same way as MOP0 and MOP1.
5. The output of DISP0 and DISP1 can be written back to the memory by WBC.

### 1.2.7. Maruko




※ Note
sRGB,TTL,B656, and BT1120 cannot be output at the same time, only one of them can be output at a time.

※ Note
The differences among different chip series are shown in the table below:

| Output interface / Chip series | HDMI | VGA | MIPI DSI | TTL | CVBS | sRGB |
|---|---|---|---|---|---|---|
| Pretzel | not support | not support | not support | support | not support | not support |
| Macaron | not support | not support | not support | not support | not support | not support |
| Taiyaki | support | support | not support | not support | not support | not support |
| Takoyaki | not support | not support | support | support | not support | not support |
| Pudding | not support | not support | not support | support | not support | not support |
| Ispahan | not support | not support | not support | not support | not support | not support |
| MSR930 | support | support | support | support | not support | not support |
| Tiramisu | support | support | support | support | support | support |
| Ikayaki | not support | not support | not support | support | not support | support |
| Muffin | support | support | support | support | support | support |
| Mochi | support | support | not support | not support | support | not support |
| Maruko | not support | not support | not support | support | not support | support |

## 1.3. Keyword

- DEV
  Device, which corresponded to DISP0/1 in Flow Chart.

- MOP
  A hardware unit that reads image data in memory and performs puzzle processing.

- LAYER
  Video layer. MOP hardware abstraction layer.

- CSC
  Color space conversion unit

- OSD Mixer
  UI overlay

- GUI FB/HW Cursor
  GOP output

- PIP
  Picture In Picture

- WBC
  Write back channel, it can capture video layer or device-level video data, used for multi-dev homologous display.

# 2. API LIST

The MI DISP module provides the following APIs:

Table 2-1: API list

| Name of API | Function |
|---|---|
| MI_DISP_Enable | Enable video output device |
| MI_DISP_Disable | Disable video output device |
| MI_DISP_SetPubAttr | Set public attribute of video output device |
| MI_DISP_GetPubAttr | Get public attribute of video output device |
| MI_DISP_DeviceAttach | Enable video output devices from the same source |
| MI_DISP_DeviceDetach | Disable video output devices from the same source |
| MI_DISP_EnableVideoLayer | Enable video layer |
| MI_DISP_DisableVideoLayer | Disable video layer |
| MI_DISP_SetVideoLayerAttr | Set video layer attribute |
| MI_DISP_GetVideoLayerAttr | Get video layer attribute |
| MI_DISP_BindVideoLayer | Bind video layer to specified device |
| MI_DISP_UnBindVideoLayer | Unbind video layer from specified device |
| MI_DISP_SetPlayToleration | Set playback toleration |
| MI_DISP_GetPlayToleration | Get playback toleration |
| MI_DISP_GetScreenFrame | Get screen output frame data |
| MI_DISP_ReleaseScreenFrame | Release screen output frame data |
| MI_DISP_SetVideoLayerAttrBatchBegin | Set the input port on the video layer related operations start |
| MI_DISP_SetVideoLayerAttrBatchEnd | Set the input port on the video layer related operations end |
| MI_DISP_EnableInputPort | Enable specified video input port |
| MI_DISP_DisableInputPort | Disable specified video input port |
| MI_DISP_SetInputPortAttr | Set specified video input port attribute |
| MI_DISP_GetInputPortAttr | Get specified video input port attribute |
| MI_DISP_SetInputPortDispPos | Set specified video input port display position |
| MI_DISP_GetInputPortDispPos | Get specified video input port display position |
| MI_DISP_PauseInputPort | Pause specified video input port |

| Name of API | Function |
|---|---|
| MI_DISP_ResumeInputPort | Resume specified video input port |
| MI_DISP_StepInputPort | Play specified video input port by single frame |
| MI_DISP_ShowInputPort | Show specified video input port |
| MI_DISP_HideInputPort | Hide specified video input port |
| MI_DISP_SetInputPortSyncMode | Set specified video input port sync mode |
| MI_DISP_QueryInputPortStat | Query video input port status |
| MI_DISP_SetZoomInWindow | Crop video input port |
| MI_DISP_GetVgaParam | Get VGA output device parameter |
| MI_DISP_SetVgaParam | Set VGA output device parameter |
| MI_DISP_GetHdmiParam | Get HDMI output device parameter |
| MI_DISP_SetHdmiParam | Set HDMI output device parameter |
| MI_DISP_GetLcdParam | Get LCD output device parameter |
| MI_DISP_SetLcdParam | Set LCD output device parameter |
| MI_DISP_GetCvbsParam | Get CVBS output device parameter |
| MI_DISP_SetCvbsParam | Set CVBS output device parameter |
| MI_DISP_DeviceGetColorTempeture | Get output image color temperature information |
| MI_DISP_DeviceSetColorTempeture | Set output image color temperature information |
| MI_DISP_DeviceSetGammaParam | Adjust output image gamma parameter |
| MI_DISP_SetVideoLayerRotateMode | Set the rotation Angle of the output image |
| MI_DISP_ClearInputPortBuffer | Clear the content of specified video input port |
| MI_DISP_InitDev | Initialize disp device |
| MI_DISP_DeInitDev | De-initialize disp device |
| MI_DISP_SetWBCSource | Set the write-back source of the write-back device |
| MI_DISP_GetWBCSource | Get the write-back source of the write-back device |
| MI_DISP_SetWBCAttr | Set write-back device attributes |
| MI_DISP_GetWBCAttr | Get write-back device attributes |
| MI_DISP_EnableWBC | Enable write-back device |
| MI_DISP_DisableWBC | Disable write-back device |

## 2.1. MI_DISP_Enable

- Function

  Enable video output device

- Syntax

  MI_S32 MI_DISP_Enable (MI_DISP_DEV DispDev);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Since system will not initialize to enable the device, the device should first be enabled before the video output function can be used.
  - Before calling the device enable function, the public attribute of the device must be set beforehand; otherwise, a device not configured error code will be returned.

- Example

```
MI_U32 DispDev = 0;
MI_U32 DispLayer = 0;
MI_U32 DispInport = 0;
MI_DISP_PubAttr_t stPubAttr;
MI_DISP_VideoLayerAttr_t stLayerAttr;
MI_DISP_InputPortAttr_t stInputPortAttr;
MI_DISP_VidWinRect_t stWinRect;
MI_DISP_RotateConfig_t stRotateConfig;

memset (&stPubAttr,0,sizeof (MI_DISP_PubAttr_t));
memset (&stLayerAttr,0,sizeof (MI_DISP_VideoLayerAttr_t));
memset (&stInputPortAttr,0,sizeof (MI_DISP_InputPortAttr_t));
memset(&stWinRect, 0, sizeof(MI_DISP_VidWinRect_t));
memset(&stRotateConfig, 0, sizeof(MI_DISP_RotateConfig_t));

stPubAttr.eIntfSync = E_MI_DISP_OUTPUT_1080P60;
stPubAttr.eIntfType = E_MI_DISP_INTF_HDMI;
MI_DISP_SetPubAttr (DispDev, &stPubAttr);
stPubAttr.eIntfType = E_MI_DISP_INTF_VGA;
MI_DISP_SetPubAttr(DispDev, &stPubAttr); // Non-essential steps
MI_DISP_Enable (DispDev);
```

```
//HDMI related init
//Necessary steps. Not listed here

stLayerAttr.stVidLayerSize.u16Width = 1920;
stLayerAttr.stVidLayerSize.u16Height = 1080;
stLayerAttr.stVidLayerDispWin.u16X = 0;
stLayerAttr.stVidLayerDispWin.u16Y = 0;
stLayerAttr.stVidLayerDispWin.u16Width = 1920;
stLayerAttr.stVidLayerDispWin.u16Height = 1080;
stRotateConfig.eRotateMode = E_MI_DISP_ROTATE_NONE;
MI_DISP_BindVideoLayer (DispLayer,DispDev);
MI_DISP_SetVideoLayerAttr (DispLayer, &stLayerAttr);
MI_DISP_EnableVideoLayer (DispLayer);
MI_DISP_SetVideoLayerRotateMode(DispLayer, &stRotateConfig); // Non-essential steps

stInputPortAttr.u16SrcWidth = 1920;
stInputPortAttr.u16SrcHeight = 1080;
stInputPortAttr.stDispWin.u16X = 0;
stInputPortAttr.stDispWin.u16Y = 0;
stInputPortAttr.stDispWin.u16Width = 1920;
stInputPortAttr.stDispWin.u16Height = 1080;
stWinRect.u16X = 0;
stWinRect.u16Y = 0;
stWinRect.u16Width = 1920;
stWinRect.u16Height = 1080;
MI_DISP_SetInputPortAttr (DispLayer, DispInport, &stInputPortAttr);
MI_DISP_SetZoomInWindow(DispLayer, DispInport, &stWinRect); // Non-essential steps
MI_DISP_SetVideoLayerAttrBatchBegin(DispLayer); // Non-essential steps
MI_DISP_EnableInputPort (DispLayer, DispInport);
MI_DISP_SetVideoLayerAttrBatchEnd(DispLayer); // Non-essential steps


//exit flow
MI_DISP_SetVideoLayerAttrBatchBegin(DispLayer); // Non-essential steps
MI_DISP_DisableInputPort (DispLayer, DispInport);
MI_DISP_SetVideoLayerAttrBatchEnd(DispLayer); // Non-essential steps
MI_DISP_DisableVideoLayer (DispLayer);
MI_DISP_UnBindVideoLayer (DispLayer, DispDev);
MI_DISP_Disable (DispDev);
```

- Related API

  MI_DISP_Disable

## 2.2. MI_DISP_Disable

- Function

  Disable video output device

- Syntax

  MI_S32  MI_DISP_Disable (MI_DISP_DEV DispDev);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  N/A.

- Example

  N/A.

- Related API

  MI_DISP_Enable

## 2.3. MI_DISP_SetPubAttr

- Function

  Set public attribute of video output device

- Syntax

  MI_S32 MI_DISP_SetPubAttr (MI_DISP_DEV DispDev, MI_DISP_PubAttr_t *pstPubAttr);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Output device number | Input |
| pstPubAttr | Pointer to output device public attribute structure | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  N/A.

- Example

  N/A.

- Related API

  MI_DISP_GetPubAttr

## 2.4. MI_DISP_GetPubAttr

- Function

  Get public attribute of video output device

- Syntax

  MI_S32 MI_DISP_GetPubAttr (MI_DISP_DEV DispDev, MI_DISP_PubAttr_t *pstPubAttr);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Output device number | Input |
| pstPubAttr | Pointer to output device public attribute structure | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  N/A.

- Example

  N/A.

- Related API

  MI_DISP_SetPubAttr

## 2.5. MI_DISP_DeviceAttach

- Function

  Enable video output devices from the same source, which can achieve the effect of same screen with different output interfaces available simultaneously.

- Syntax

  MI_S32 MI_DISP_DeviceAttach (MI_DISP_DEV DispSrcDev, MI_DISP_DEV DispDstDev, const

MI_DISP_DevAttachAttr_t *pstAttachAttr);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispSrcDev | Output device number (video source device) | Input |
| DispDstDev | Output device number (video destination device) | Input |
| pstAttachAttr | Video destination output attributes | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - This API only supports chips featuring two video devices.
  - Currently supported chips include MSR930 and MSR650x.

- Example

  N/A.

- Related API

  MI_DISP_DeviceDetach

## 2.6. MI_DISP_DeviceDetach

- Function

  Disable video output devices from the same source

- Syntax

  MI_S32 MI_DISP_DeviceDetach (MI_DISP_DEV DispSrcDev, MI_DISP_DEV DispDstDev);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispSrcDev | Output device number (video source device) | Input |
| DispDstDev | Output device number (video destination device) | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement

- Header file: mi_disp.h, mi_disp_datatype.h
- Lib: libmi_disp.a / libmi_disp.so

➢ Note

- This API only supports chips featuring two video devices.
- Currently supported chips include MSR930 and MSR650x.

➢ Example

N/A.

➢ Related API

MI_DISP_DeviceAttach

## 2.7. MI_DISP_EnableVideoLayer

➢ Function

Enable video layer

➢ Syntax

MI_S32 MI_DISP_EnableVideoLayer (MI_DISP_LAYER DispLayer);

➢ Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |

➢ Return Value

- Zero: Successful
- Non-zero: Failed, see error code for details

➢ Requirement

- Header file: mi_disp.h, mi_disp_datatype.h
- Lib: libmi_disp.a / libmi_disp.so

➢ Note

N/A.

➢ Example

N/A.

➢ Related API

MI_DISP_DisableVideoLayer

## 2.8. MI_DISP_DisableVideoLayer

➢ Function

Disable video layer

- Syntax

  MI_S32 MI_DISP_DisableVideoLayer (MI_DISP_LAYER DispLayer);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Before calling this function, make sure all input port of video layer has been disabled.

- Example

  N/A.

- Related API

  MI_DISP_EnableVideoLayer

## 2.9. MI_DISP_SetVideoLayerAttr

- Function

  Set video layer attribute

- Syntax

  MI_S32 MI_DISP_SetVideoLayerAttr (MI_DISP_LAYER DispLayer, MI_DISP_VideoLayerAttr_t *pstLayerAttr);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| pstLayerAttr | Pointer to video layer attribute structure | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note

  N/A.
- Example

  N/A.
- Related API

  MI_DISP_GetVideoLayerAttr

## 2.10. MI_DISP_GetVideoLayerAttr

- Function

  Get video layer attribute
- Syntax

  MI_S32 MI_DISP_GetVideoLayerAttr (MI_DISP_LAYER DispLayer, MI_DISP_VideoLayerAttr_t *pstLayerAttr);
- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| pstLayerAttr | Pointer to video layer attribute structure | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note

  N/A.
- Example

  N/A.
- Related API

  MI_DISP_SetVideoLayerAttr

## 2.11. MI_DISP_BindVideoLayer

- Function

  Bind video layer to specified device

- Syntax

  MI_S32 MI_DISP_BindVideoLayer (MI_DISP_LAYER DispLayer, MI_DISP_DEV DispDev);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| DispDev | Output device number | Input |

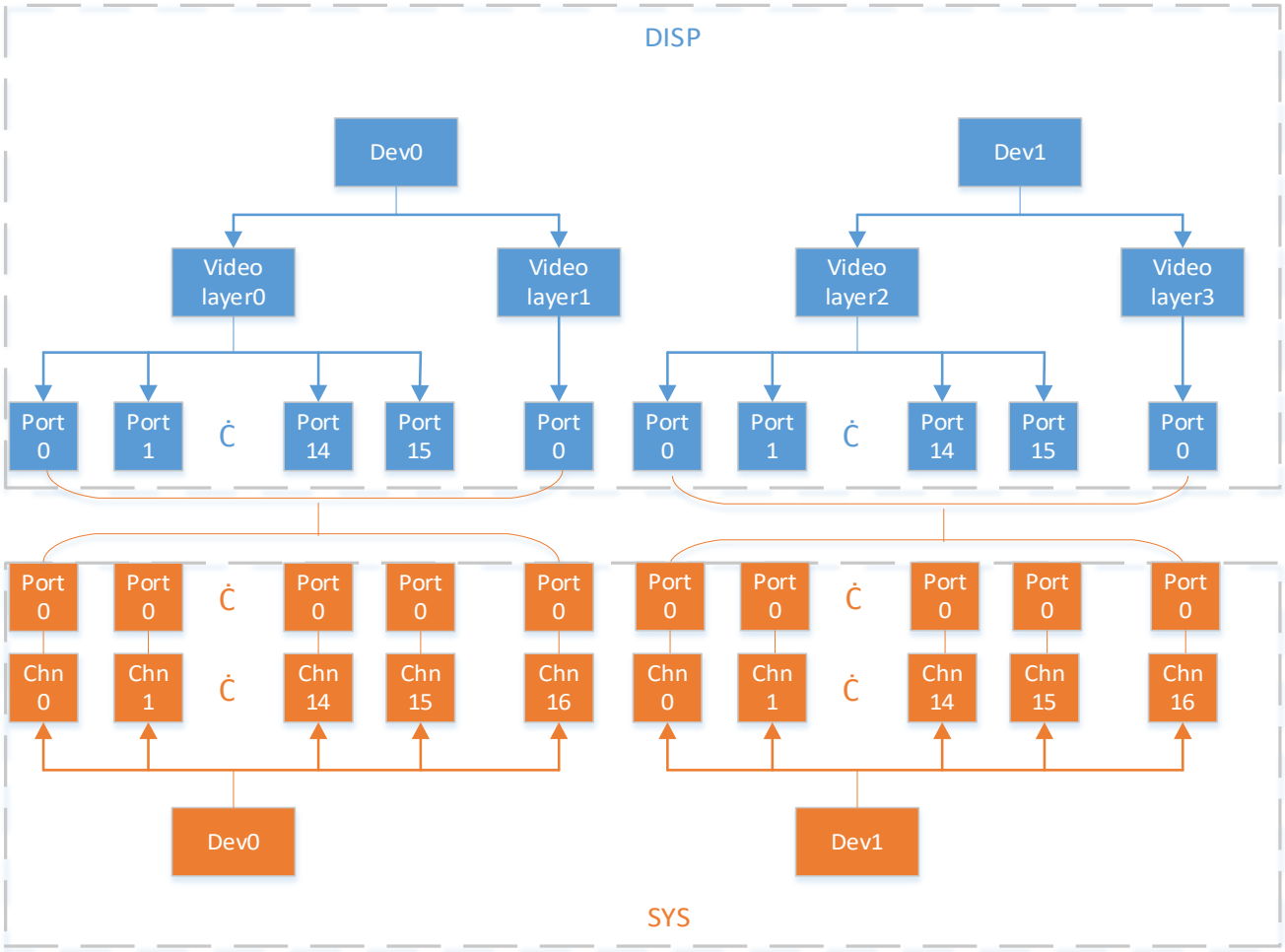
- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

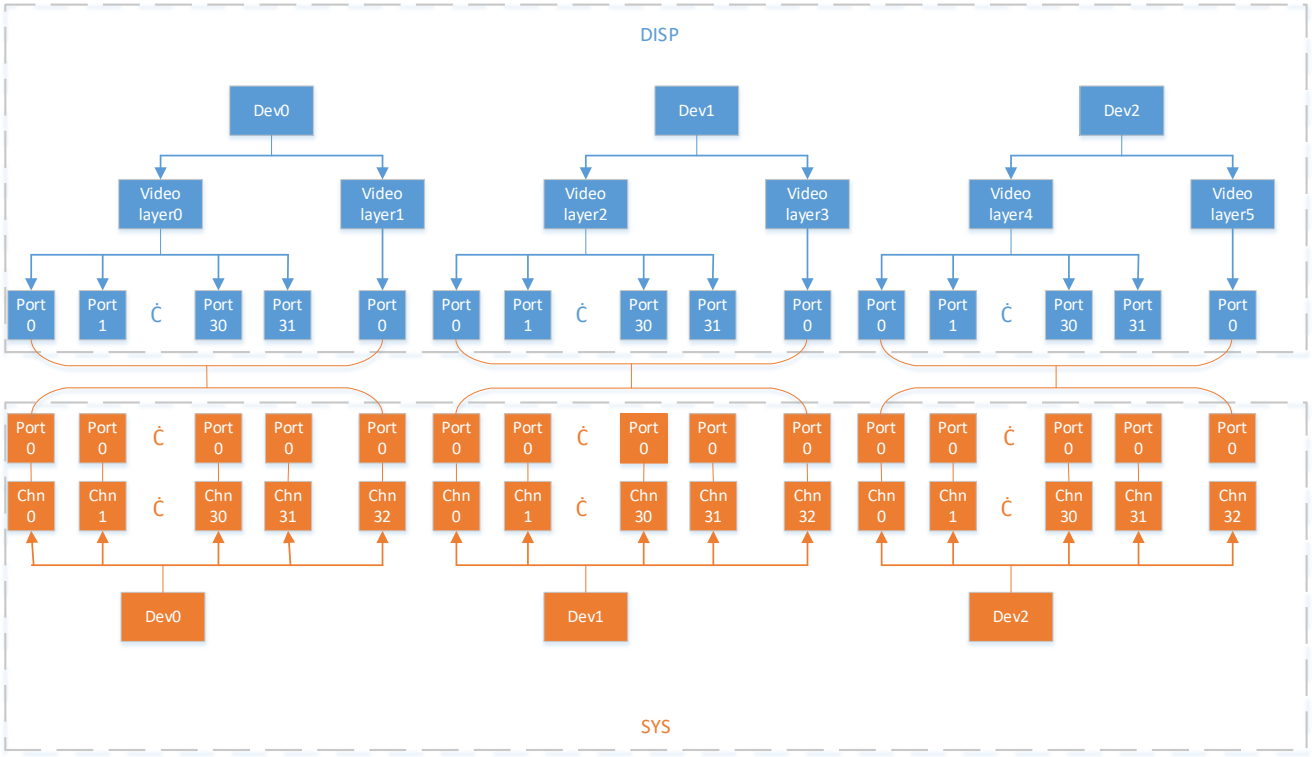
➢ Note

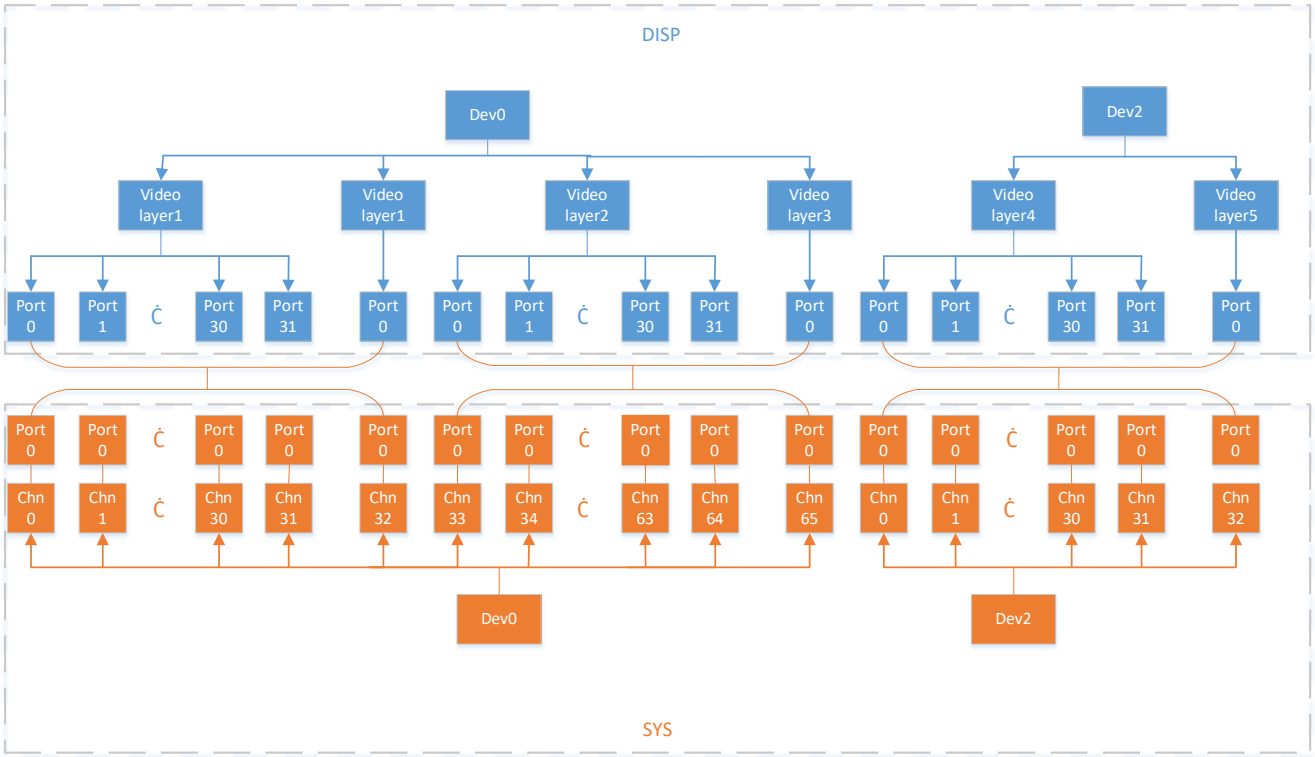
- Before calling this function, make sure the all input port of video layer has been disabled
- When DISP is bound to other modules, the mapping relationship between device/video layer/input port of DISP and the corresponding channel/port are as follows.

1）Tiramisu

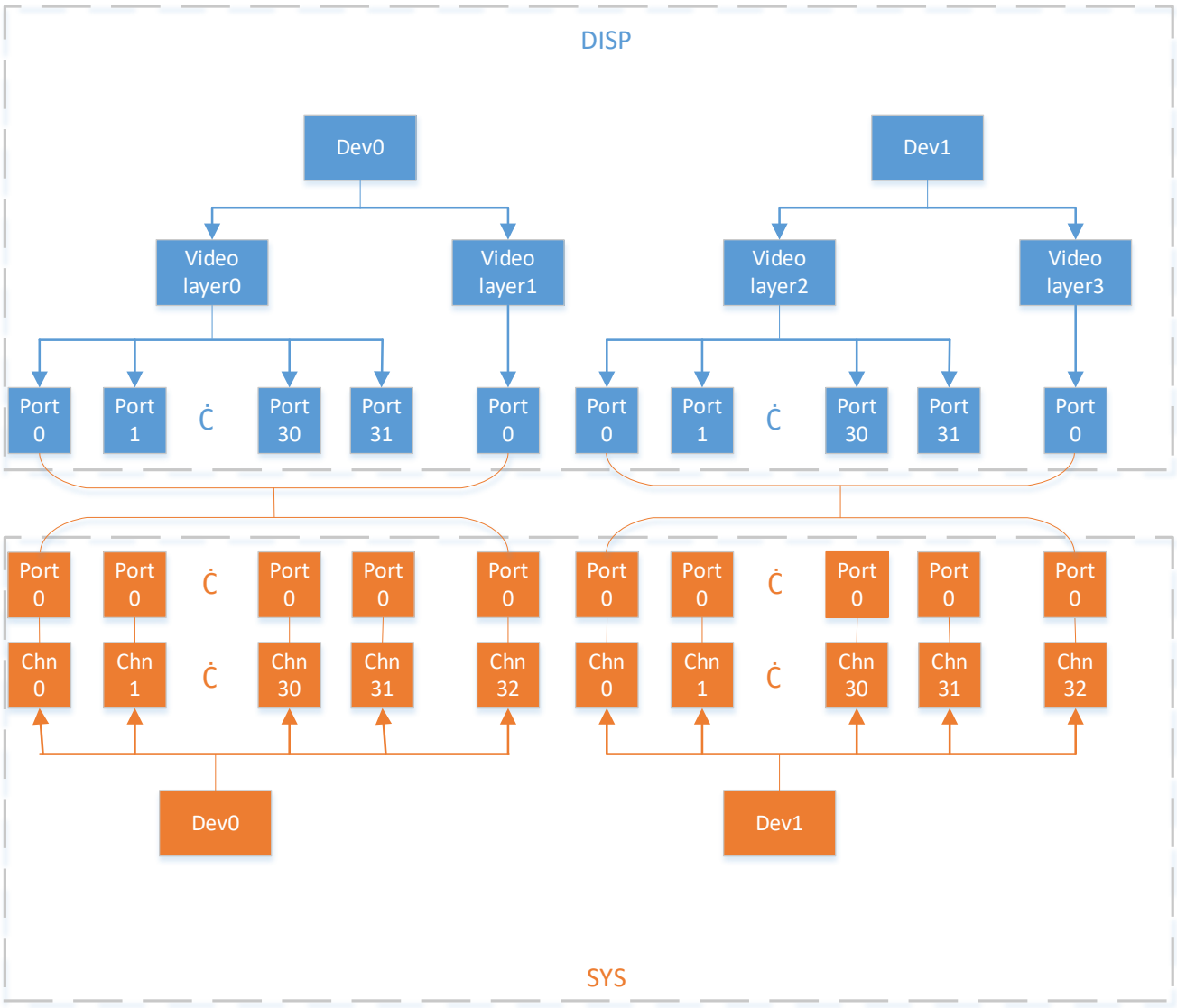


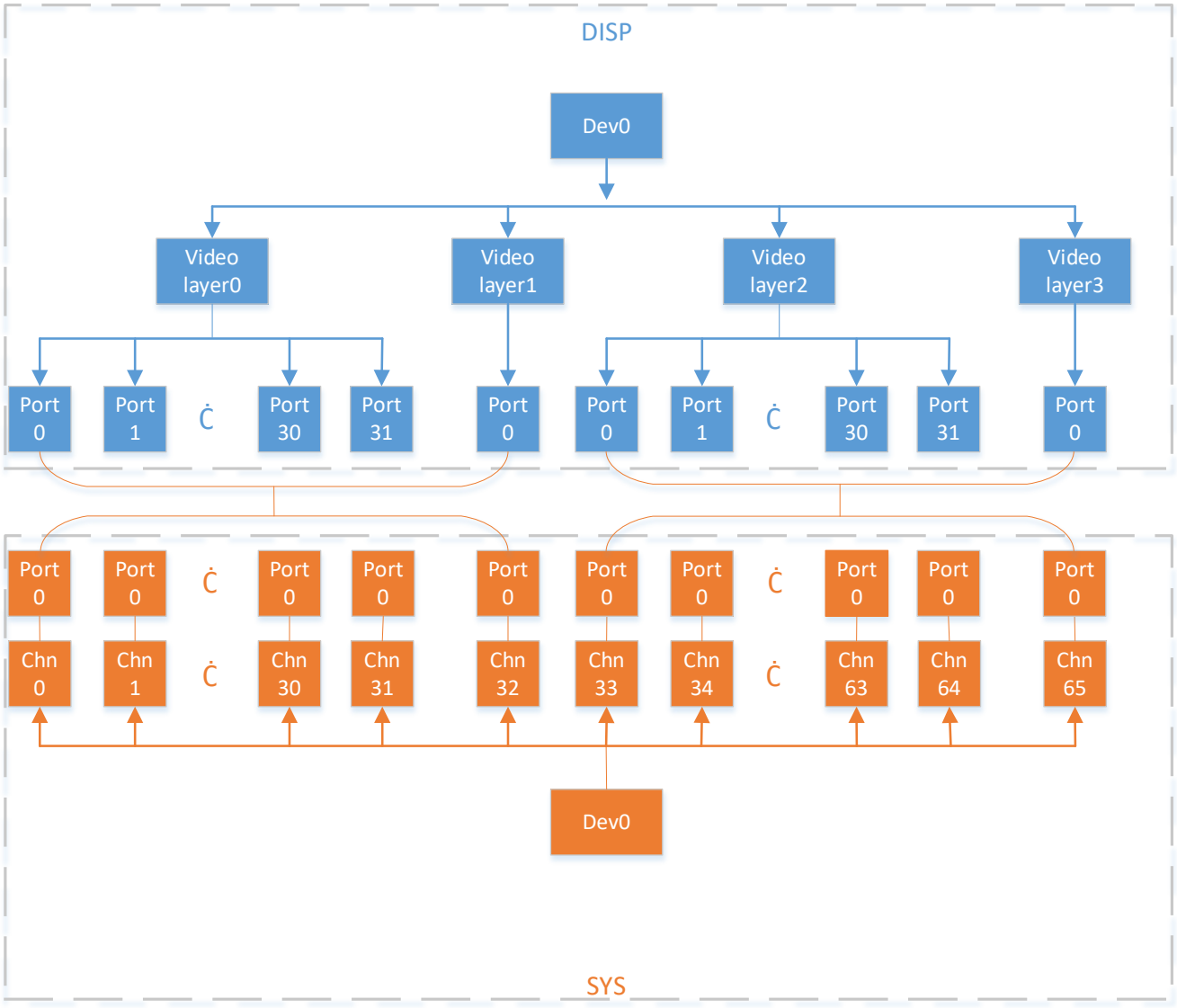
2) Muffin supports two binding relationships



3) Mochi supports two binding relationships






- Example

  N/A.

- Related API

  MI_DISP_UnBindVideoLayer

## 2.12. MI_DISP_UnBindVideoLayer

- Function

  Unbind video layer from specified device

- Syntax

  MI_S32 MI_DISP_UnBindVideoLayer (MI_DISP_LAYER DispLayer, MI_DISP_DEV DispDev);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| DispDev | Output device number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Before calling this function, make sure all input port of video layer has been disabled.
  - Before calling this function, make sure video layer has been disabled

- Example

  N/A.

- Related API

  MI_DISP_BindVideoLayer

## 2.13. MI_DISP_SetPlayToleration

- Function

  Set playback toleration

- Syntax

  MI_S32 MI_DISP_SetPlayToleration (MI_DISP_LAYER DispLayer, MI_U32 u32Toleration);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| u32Toleration | Playback toleration used to adjust the error tolerance of the frame rate control algorithm | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Playback toleration is measured in unit of millisecond.
  - Before calling this function, make sure the video layer has been enabled.
  - Currently supported chips include MSR930 and MSR650x.

- Example

  N/A.

- Related API

  MI_DISP_GetPlayToleration

## 2.14. MI_DISP_GetPlayToleration

- Function

  Get playback toleration

- Syntax

  MI_S32 MI_DISP_GetPlayToleration (MI_DISP_LAYER DispLayer, MI_U32 *pu32Toleration);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| pu32Toleration | Playback toleration. | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Playback toleration is measured in unit of millisecond.
  - Before calling this function, make sure the video layer has been enabled.
  - Currently supported chips include MSR930 and MSR650x.

- Example

  N/A.

- Related API

  MI_DISP_SetPlayToleration

## 2.15. MI_DISP_GetScreenFrame

- Function

  Get output screen frame data

- Syntax

  MI_S32 MI_DISP_GetScreenFrame (MI_DISP_LAYER DispLayer, MI_DISP_VideoFrame_t *pstVFrame, MI_S32 s32MilliSec);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| pstVFrame | Pointer to the gotten output screen frame data structure | Output |
| s32MilliSec | Block the port when the timeout parameter s32MilliSec is set to -1; do not block the port when the timeout parameter s32MilliSec is set to 0. Value larger than 0 represents the timeout waiting time. The timeout waiting time is in unit of millisecond (ms). | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Before calling this function, make sure the video layer has been enabled.
  - Currently supported chips include MSR930 and MSR650x.

- Example

  N/A.

- Related API

  MI_DISP_ReleaseScreenFrame

## 2.16. MI_DISP_ReleaseScreenFrame

- Function

  Release output screen frame data

- Syntax

MI_S32 MI_DISP_ReleaseScreenFrame (MI_DISP_LAYER DispLayer, MI_DISP_VideoFrame_t *pstVFrame);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| pstVFrame | Pointer to the gotten output screen frame data structure | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Before calling this function, make sure the video layer has been enabled.
  - Currently supported chips include MSR930 and MSR650x.
- Example

  N/A.

- Related API

  MI_DISP_GetScreenFrame

## 2.17. MI_DISP_SetVideoLayerAttrBatchBegin

- Function

  Set the input port on the video layer related operations start.

- Syntax

  MI_S32 MI_DISP_SetVideoLayerAttrBatchBegin(MI_DISP_LAYER DispLayer);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h

- Lib: libmi_disp.a / libmi_disp.so

➢ Note

- Before calling this function, make sure the video layer has been enabled.
- The MI_DISP_SetVideoLayerAttrBatchBegin and MI_DISP_SetVideoLayerAttrBatchEnd must be paired, otherwise the related operations after MI_DISP_SetVideoLayerAttrBatchBegin will not take effect.
- This interface supports batch processing of the following interfaces: MI_DISP_EnableInputPort, MI_DISP_DisableInputPort, MI_DISP_HideInputPort, MI_DISP_ShowInputPort.
- It will take effect immediately for interfaces that do not support batch processing.
- Input port operations in batch processing are all effective after MI_DISP_SetVideoLayerAttrBatchEnd.
  For example, batch processing disables input port 0~15, interface order is MI_DISP_SetVideoLayerAttrBatchBegin, MI_DISP_DisableInputPort (0~15), MI_DISP_SetVideoLayerAttrBatchEnd.
- Currently supported chips include Muffin and Mochi.

➢ Example

N/A.

➢ Related API

MI_DISP_SetVideoLayerAttrBatchEnd

## 2.18. MI_DISP_SetVideoLayerAttrBatchEnd

➢ Function

Set the input port on the video layer related operations end.

➢ Syntax

MI_S32 MI_DISP_SetVideoLayerAttrBatchEnd(MI_DISP_LAYER DispLayer);

➢ Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |

➢ Return Value

- Zero: Successful
- Non-zero: Failed, see error code for details

➢ Requirement

- Header file: mi_disp.h, mi_disp_datatype.h
- Lib: libmi_disp.a / libmi_disp.so

➢ Note

- Before calling this function, make sure the MI_DISP_SetVideoLayerAttrBatchBegin has been called.
- Before calling this function, make sure the video layer has been enabled.
- Currently supported chips include Muffin and Mochi.

➢ Example

N/A.

➢ Related API

MI_DISP_SetVideoLayerAttrBatchBegin

## 2.19. MI_DISP_EnableInputPort

➢ Function

Enable specified video input port

➢ Syntax

MI_S32 MI_DISP_EnableInputPort (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort);

➢ Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |

➢ Return Value

- Zero: Successful
- Non-zero: Failed, see error code for details

➢ Requirement

- Header file: mi_disp.h, mi_disp_datatype.h
- Lib: libmi_disp.a / libmi_disp.so

➢ Note

- Before calling this function, make sure the video layer has been enabled.
- Before calling this function, make sure video layer has been bound

➢ Example

N/A.

➢ Related API

MI_DISP_DisableInputPort

## 2.20. MI_DISP_DisableInputPort

- Function

  Disable specified video input port

- Syntax

  MI_S32 MI_DISP_DisableInputPort (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A.

- Related API

  MI_DISP_EnableInputPort

## 2.21. MI_DISP_SetInputPortAttr

- Function

  Set specified video input port attribute

- Syntax

  MI_S32 MI_DISP_SetInputPortAttr (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort, MI_DISP_InputPortAttr_t* pstInputPortAttr);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |
| pstInputPortAttr | Video input port attribute pointer | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note
  - Before calling this function, make sure the video layer has been enabled.
  - Before calling this function, make sure video layer has been bound .
- Example

  N/A.
- Related API

  MI_DISP_GetInputPortAttr

## 2.22. MI_DISP_GetInputPortAttr

- Function

  Get specified video input port attribute
- Syntax

  MI_S32 MI_DISP_GetInputPortAttr (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort, MI_DISP_InputPortAttr_t *pstInputPortAttr);
- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |
| pstInputPortAttr | Video input port attribute pointer | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note

- Before calling this function, make sure the video layer has been enabled.
- Before calling this function, make sure video layer has been bound

➢ Example

N/A.

➢ Related API

MI_DISP_SetInputPortAttr

## 2.23. MI_DISP_SetInputPortDispPos

➢ Function

Set specified video input port display position

➢ Syntax

MI_S32 MI_DISP_SetInputPortDispPos (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort, const MI_DISP_Position_t *pstDispPos);

➢ Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |
| pstDispPos | Input port position pointer | Input |

➢ Return Value

- Zero: Successful
- Non-zero: Failed, see error code for details

➢ Requirement

- Header file: mi_disp.h, mi_disp_datatype.h
- Lib: libmi_disp.a / libmi_disp.so

➢ Note

Before calling this function, make sure the video layer has been enabled.

➢ Example

N/A.

➢ Related API

MI_DISP_GetInputPortDispPos

## 2.24. MI_DISP_GetInputPortDispPos

➢ Function

Get specified video input port display position

- Syntax

  MI_S32 MI_DISP_GetInputPortDispPos
  (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort, MI_DISP_Position_t *pstDispPos);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |
| pstDispPos | Input port position pointer | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A.

- Related API

  MI_DISP_SetInputPortDispPos

## 2.25. MI_DISP_PauseInputPort

- Function

  Pause specified video input port

- Syntax

  MI_S32 MI_DISP_PauseInputPort (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A.

- Related API

  MI_DISP_DisableInputPort
  MI_DISP_PauseInputPort
  MI_DISP_ResumeInputPort
  MI_DISP_StepInputPort
  MI_DISP_ShowInputPort
  MI_DISP_HideInputPort

## 2.26. MI_DISP_ResumeInputPort

- Function

  Resume specified video input port

- Syntax

  MI_S32 MI_DISP_ResumeInputPort (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A.

- Related API

  MI_DISP_DisableInputPort
  MI_DISP_PauseInputPort
  MI_DISP_ResumeInputPort
  MI_DISP_StepInputPort
  MI_DISP_ShowInputPort
  MI_DISP_HideInputPort

## 2.27. MI_DISP_StepInputPort

- Function

  Play specified video input port by single frame

- Syntax

  MI_S32 MI_DISP_StepInputPort (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Before calling this function, make sure the video layer has been enabled.
  - Currently supported chips include MSR930 and MSR650x.

- Example

  N/A.

- Related API

MI_DISP_DisableInputPort
MI_DISP_PauseInputPort
MI_DISP_ResumeInputPort
MI_DISP_StepInputPort
MI_DISP_ShowInputPort
MI_DISP_HideInputPort

## 2.28. MI_DISP_ShowInputPort

- Function

  Show specified video input port

- Syntax

  MI_S32 MI_DISP_ShowInputPort (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A.

- Related API

  MI_DISP_DisableInputPort
  MI_DISP_PauseInputPort
  MI_DISP_ResumeInputPort
  MI_DISP_StepInputPort
  MI_DISP_ShowInputPort
  MI_DISP_HideInputPort

## 2.29. MI_DISP_HideInputPort

- Function

  Hide specified video input port

- Syntax

  MI_S32 MI_DISP_HideInputPort (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A.

- Related API

  MI_DISP_DisableInputPort
  MI_DISP_PauseInputPort
  MI_DISP_ResumeInputPort
  MI_DISP_StepInputPort
  MI_DISP_ShowInputPort
  MI_DISP_HideInputPort

## 2.30. MI_DISP_SetInputPortSyncMode

- Function

  Set specified video input port sync mode

- Syntax

  MI_S32 MI_DISP_SetInputPortSyncMode (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort, MI_DISP_SyncMode_e eMode);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |
| eMode | Input port sync mode | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note
  - Before calling this function, make sure the video layer has been enabled.
  - Currently supported chips include MSR930 and MSR650x.
- Example

  N/A.
- Related API

  N/A.

## 2.31. MI_DISP_QueryInputPortStat

- Function

  Query video input port status
- Syntax

  MI_S32 MI_DISP_QueryInputPortStat (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort, MI_DISP_QueryChanelStatus_t *pstStatus);
- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output display layer number | Input |
| LayerInputPort | Video input port number | Input |
| pstStatus | Input port status pointer | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement

- Header file: mi_disp.h, mi_disp_datatype.h
- Lib: libmi_disp.a / libmi_disp.so

➢ Note

Before calling this function, make sure the video layer has been enabled.

➢ Example

N/A.

➢ Related API

N/A.

## 2.32. MI_DISP_SetZoomInWindow

➢ Function

Crop video input port.

➢ Syntax

MI_S32 MI_DISP_SetZoomInWindow (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort, MI_DISP_VidWinRect_t* pstZoomRect);

➢ Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output video layer number. | Input |
| LayerInputPort | Video input port number. | Input |
| pstZoomRect | Video cropping attribute pointer. | Input |

➢ Return Value

- Zero: Successful
- Non-zero: Failed, see error code for details

➢ Requirement

- Header file: mi_disp.h, mi_disp_datatype.h
- Lib: libmi_disp.a / libmi_disp.so

➢ Note

The video layer must be enabled before calling.

➢ Example

N/A.

➢ Related API

MI_DISP_VidWinRect_t

## 2.33. MI_DISP_GetVgaParam

- Function

  Get VGA output device parameter

- Syntax

  MI_S32 MI_DISP_GetVgaParam (MI_DISP_DEV DispDev, MI_DISP_VgaParam_t *pstVgaParam);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |
| pstVgaParam | Pointer to VGA image output effect structure | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A.

- Related API

  MI_DISP_SetVgaParam

## 2.34. MI_DISP_SetVgaParam

- Function

  Set VGA output device parameter

- Syntax

  MI_S32 MI_DISP_SetVgaParam (MI_DISP_DEV DispDev, MI_DISP_VgaParam_t *pstVgaParam);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |
| pstVgaParam | Pointer to VGA image output effect structure | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note

  Before calling this function, make sure the video layer has been enabled.
- Example

  N/A.
- Related API

  MI_DISP_GetVgaParam

## 2.35. MI_DISP_GetHdmiParam

- Function

  Get HDMI output device parameter
- Syntax

  MI_S32 MI_DISP_GetHdmiParam (MI_DISP_DEV DispDev, MI_DISP_HdmiParam_t *pstHdmiParam);
- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video Output device number | Input |
| pstHdmiParam | Pointer to HDMI image output effect structure | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note

  Before calling this function, make sure the video layer has been enabled.
- Example

  N/A.
- Related API

MI_DISP_SetHdmiParam

## 2.36. MI_DISP_SetHdmiParam

- Function

  Set HDMI output device parameter

- Syntax

  MI_S32 MI_DISP_SetHdmiParam (MI_DISP_DEV DispDev, MI_DISP_HdmiParam_t *pstHdmiParam);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |
| pstHdmiParam | Pointer to HDMI image output effect structure | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A.

- Related API

  MI_DISP_GetHdmiParam

## 2.37. MI_DISP_GetLcdParam

- Function

  Get LCD output device parameter

- Syntax

  MI_S32 MI_DISP_GetLcdParam (MI_DISP_DEV DispDev, MI_DISP_LcdParam_t *pstLcdParam);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|

| | | |
|---|---|---|
| DispDev | Video output device number | Input |
| pstLcdParam | Pointer to LCD image output effect structure | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note

  Before calling this function, make sure the video layer has been enabled.
- Example

  N/A
- Related API

  MI_DISP_SetLcdParam

## 2.38. MI_DISP_SetLcdParam

- Function

  Set LCD output device parameter
- Syntax

  MI_S32 MI_DISP_SetLcdParam (MI_DISP_DEV DispDev, MI_DISP_LcdParam_t *pstLcdParam);
- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |
| pstLcdParam | Pointer to LCD image output effect structure | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A

- Related API

  MI_DISP_GetLcdParam

## 2.39. MI_DISP_GetCvbsParam

- Function

  Get CVBS output device parameter

- Syntax

  MI_S32 MI_DISP_GetCvbsParam(MI_DISP_DEV DispDev, MI_DISP_CvbsParam_t *pstCvbsParam);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |
| pstCvbsParam | CVBS image output effect structure pointer | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A

- Related API

  MI_DISP_SetCvbsParam

## 2.40. MI_DISP_SetCvbsParam

- Function

  Set CVBS output device parameter

- Syntax

  MI_S32 MI_DISP_SetCvbsParam(MI_DISP_DEV DispDev, MI_DISP_CvbsParam_t *pstCvbsParam);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |
| pstCvbsParam | CVBS image output effect structure pointer | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A

- Related API

  MI_DISP_GetCvbsParam

## 2.41. MI_DISP_DeviceGetColorTempeture

- Function

  Get output image color temperature information

- Syntax

  MI_S32 MI_DISP_DeviceGetColorTempeture (MI_DISP_DEV DispDev, MI_DISP_ColorTemperature_t *pstColorTempInfo);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |
| pstColorTempInfo | Pointer to output image color temperature structure | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

- Before calling this function, make sure the video layer has been enabled.
- Currently supported chips include Takoyaki series.

➢ Example

N/A

➢ Related API

MI_DISP_DeviceSetColorTempeture

## 2.42. MI_DISP_DeviceSetColorTempeture

➢ Function

Set image output color temperature information

➢ Syntax

MI_S32 MI_DISP_DeviceSetColorTempeture (MI_DISP_DEV DispDev, MI_DISP_ColorTemperature_t *pstColorTempInfo);

➢ Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |
| pstColorTempInfo | Pointer to output image color temperature structure | Input |

➢ Return Value

- Zero: Successful
- Non-zero: Failed, see error code for details

➢ Requirement

- Header file: mi_disp.h, mi_disp_datatype.h
- Lib: libmi_disp.a / libmi_disp.so

➢ Note

- Before calling this function, make sure the video layer has been enabled.
- Currently supported chips include Takoyaki series.

➢ Example

N/A

➢ Related API

MI_DISP_DeviceGetColorTempeture

## 2.43. MI_DISP_DeviceSetGammaParam

➢ Function

Adjust output image Gamma

- Syntax

  MI_S32 MI_DISP_DeviceSetGammaParam (MI_DISP_DEV DispDev, MI_DISP_GammaParam_t * pstGammaParam);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispDev | Video output device number | Input |
| pstGammaParam | Pointer to output image gamma structure | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Before calling this function, make sure the video layer has been enabled.
  - Currently supported chips include Takoyaki series.

- Example

  N/A

- Related API

  N/A

## 2.44. MI_DISP_SetVideoLayerRotateMode

- Function

  Set video layer rotation mode

- Syntax

  MI_S32 MI_DISP_SetVideoLayerRotateMode (MI_DISP_LAYER DispLayer, MI_DISP_RotateConfig_t *pstRotateConfig);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output layer number | Input |
| pstRotateConfig | Pointer to rotation angle structure of the output image | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Currently supported chips include Takoyaki/Tiramisu/Muffin series. Only Takoyaki support 180 degrees of rotation.
  - Before calling this function, make sure video layer has been enabled.
  - Before calling this function, make sure all input port of video layer has been disabled.
  - Need to be bound by VDEC/SCL.
  - Each video layer only supports one input port. For example, when turn on 90 degrees of rotation, each video layer can only enable input port 0.

- Example

  N/A

- Related API

  N/A

## 2.45. MI_DISP_ClearInputPortBuffer

- Function

  Clear the content of specified video input port

- Syntax

  MI_S32 MI_DISP_ClearInputPortBuffer (MI_DISP_LAYER DispLayer, MI_DISP_INPUTPORT LayerInputPort, MI_BOOL bClrAll);

- Parameter

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispLayer | Video output layer number | Input |
| LayerInputPort | Video input port number | Input |
| bClrAll | The flag of clearing current display content | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  Before calling this function, make sure the video layer has been enabled.

- Example

  N/A

- Related API

  N/A

## 2.46. MI_DISP_InitDev

- Function

  Initialize disp device.

- Syntax

  MI_S32 MI_DISP_InitDev (MI_DISP_InitParam_t *pstInitParam);

- Parameters

| Parameter Name | Description | Input/Output |
|---|---|---|
| pstInitParam | Initialization Parameter<br>Currently unused and can be configured as NULL | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  If this interface is not called before calling any DISP API, the device will be initialized automatically in the code.

- Example

  N/A

- Related API

  N/A

## 2.47. MI_DISP_DeInitDev

- Function

  De-initialize disp device.

- Syntax

MI_S32 MI_DISP_DeInitDev (void);

- Parameters

  N/A

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - This interface should be called after the device has been initialized; otherwise, a failed message will be returned.
  - If this interface is not called after the app exited, the disp device will be auto de-initialized.

- Example

  N/A

- Related API

  N/A

## 2.48. MI_DISP_SetWBCSource

- Function

  Set the write-back source of the video write-back device, the device or the video layer of the device can be set.

- Syntax

  MI_S32 MI_DISP_SetWBCSource (MI_DISP_WBC DispWbc, const MI_DISP_WBC_Source_t *pstWbcSource);

- Parameters

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispWbc | Write back the device number. | Input |
| pstWbcSource | Write back the source structure pointer. | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note
  - Only supports Tiramisu/ Muffin/ Mochi, and one write-back device.
  - u32SourceId only supported 0/1.

- Example

```
MI_DISP_WBC_Source_t stWbcSource;
MI_DISP_WBC_Attr_t stWbcAttr;

stWbcSource.eSourceType = MI_DISP_WBC_SOURCE_DEV;
stWbcSource.u32SourceId = 0;
MI_DISP_SetWBCSource (0, &stWbcSource);

stWbcAttr.stTargetSize.u32Width = 720;
stWbcAttr.stTargetSize.u32Height = 576;
stWbcAttr.ePixFormat = E_MI_SYS_PIXEL_FRAME_YUV_SEMIPLANAR_420;
MI_DISP_SetWBCAttr (0, &stWbcAttr);
MI_DISP_EnableWBC (0);

stSrcChnPort.eModId = E_MI_MODULE_ID_WBC;
stSrcChnPort.u32DevId = 0;
stSrcChnPort.u32ChnId = 0;
stSrcChnPort.u32PortId = 0;

stDstChnPort.eModId = E_MI_MODULE_ID_DISP;
stDstChnPort.u32DevId = 1;
stDstChnPort.u32ChnId = 0;
stDstChnPort.u32PortId = 0;

MI_SYS_BindChnPort (0, &stSrcChnPort, &stDstChnPort, 30, 30);
```

- Related API

  MI_DISP_GetWBCSource
  MI_DISP_SetWBCAttr
  MI_DISP_GetWBCAttr
  MI_DISP_EnableWBC
  MI_DISP_DisableWBC

## 2.49. MI_DISP_GetWBCSource

- Function

  Get the write-back source from the video write-back device.

- Syntax

  MI_S32 MI_DISP_GetWBCSource (MI_DISP_WBC DispWbc, MI_DISP_WBC_Source_t *pstWbcSource);

- Parameters

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispWbc | Write back the device number. | Input |
| pstWbcSource | Write back the source structure pointer. | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  N/A

- Example

  N/A

- Related API

  MI_DISP_SetWBCSource
  MI_DISP_SetWBCAttr
  MI_DISP_GetWBCAttr
  MI_DISP_EnableWBC
  MI_DISP_DisableWBC

## 2.50. MI_DISP_SetWBCAttr

- Function

  Set video write-back device attributes.

- Syntax

  MI_S32 MI_DISP_SetWBCAttr (MI_DISP_WBC DispWbc, const MI_DISP_WBC_Attr_t *pstWbcAttr);

- Parameters

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispWbc | Write back the device number. | Input |
| pstWbcAttr | Write back the device attribute structure pointer. | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  N/A

- Example

  N/A

- Related API

  MI_DISP_SetWBCSource
  MI_DISP_GetWBCSource
  MI_DISP_GetWBCAttr
  MI_DISP_EnableWBC
  MI_DISP_DisableWBC

## 2.51. MI_DISP_GetWBCAttr

- Function

  Get video write-back device attributes.

- Syntax

  MI_S32 MI_DISP_GetWBCAttr (MI_DISP_WBC DispWbc, MI_DISP_WBC_Attr_t *pstWbcAttr);

- Parameters

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispWbc | Write back the device number. | Input |
| pstWbcAttr | Write back the device attribute structure pointer. | Output |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note

  N/A
- Example

  N/A
- Related API

  MI_DISP_SetWBCSource
  MI_DISP_GetWBCSource
  MI_DISP_SetWBCAttr
  MI_DISP_EnableWBC
  MI_DISP_DisableWBC

## 2.52. MI_DISP_EnableWBC

- Function

  Enable video write-back device.
- Syntax

  MI_S32 MI_DISP_EnableWBC (MI_DISP_WBC DispWbc);
- Parameters

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispWbc | Write back the device number. | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details
- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so
- Note

  N/A
- Example

  N/A
- Related API

  MI_DISP_SetWBCSource
  MI_DISP_GetWBCSource

MI_DISP_SetWBCAttr
MI_DISP_EnableWBC
MI_DISP_DisableWBC

## 2.53. MI_DISP_DisableWBC

- Function

  Disable video write-back device.

- Syntax

  MI_S32 MI_DISP_DisableWBC (MI_DISP_WBC DispWbc);

- Parameters

| Parameter Name | Description | Input/Output |
|---|---|---|
| DispWbc | Write back the device number. | Input |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see error code for details

- Requirement
  - Header file: mi_disp.h, mi_disp_datatype.h
  - Lib: libmi_disp.a / libmi_disp.so

- Note

  N/A

- Example

  N/A

- Related API

  MI_DISP_SetWBCSource
  MI_DISP_GetWBCSource
  MI_DISP_SetWBCAttr
  MI_DISP_GetWBCAttr
  MI_DISP_EnableWBC

# 3. DISP DATA TYPE

The table below lists the data type definitions of the DISP module:

Table 3-2: DISP data type

| Data Structure | Description |
|---|---|
| MI_DISP_DEV | Define the DISP_DEV type |
| MI_DISP_LAYER | Define the DISP_LAYER type |
| MI_DISP_INPUTPORT | Define the LAYER_INPUTPORT type |
| MI_DISP_Interface_e | Define the DISP interface type |
| MI_DISP_IntfSync_e | Define DISP timing type |
| MI_DISP_SyncInfo_t | Define the DISP sync information |
| MI_DISP_PubAttr_t | Define the DISP public attribute |
| MI_DISP_DevAttachMode_e | Define DISP video destination output mode |
| MI_DISP_DevAttachAttr_t | Define DISP video destination output attribute |
| MI_DISP_Csc_t | Define the DISP color space conversion information |
| MI_DISP_CscMattrix_e | Define the DISP color space conversion matrix |
| MI_DISP_VgaParam_t | Define the DISP VGA parameter information |
| MI_DISP_HdmiParam_t | Define the DISP HDMI parameter information |
| MI_DISP_LcdParam_t | Define the DISP LCD parameter information |
| MI_DISP_CvbsParam_t | Define the DISP CVBS output image parameter |
| MI_DISP_ColorTemperature_t | Define the DISP color temperature parameter |
| MI_DISP_GammaParam_t | Define the DISP Gamma parameter |
| MI_DISP_VideoLayerSize_t | Define the DISP video layer attribute |
| MI_DISP_RotateMode_e | Define the DISP rotation angle information |
| MI_DISP_RotateConfig_t | Define the DISP rotation parameter |
| MI_DISP_VidWinRect_t | Define the DISP window attribute |
| MI_DISP_InputPortAttr_t | Define the DISP input port attribute |
| MI_DISP_Position_t | Define the DISP input port position |
| MI_DISP_SyncMode_e | Define the DISP sync mode |
| MI_DISP_InputPortStatus_e | Define the DISP input port status type |
| MI_DISP_QueryChannelStatus_t | Define the DISP input port status information |
| MI_DISP_VideoFrame_t | Define the DISP video frame information |

| Data Structure | Description |
|---|---|
| MI_DISP_InitParam_t | Define the initialized parameter of DISP device |
| MI_DISP_WBC | Define the type of DISP_WBC. |
| MI_DISP_WBC_SourceType_e | The source type of the data captured by the video write-back device |
| MI_DISP_WBC_Source_t | The write-back source of video write-back device |
| MI_DISP_WBC_TargetSize_t | The target size of the video write-back device |
| MI_DISP_WBC_Attr_t | Video write-back device attributes |

## 3.1. MI_DISP_DEV

- Description

  Define the DISP_DEV type

- Definition

  typedef MI_S32  MI_DISP_DEV;

- Member

  N/A.

- Note

  N/A.

## 3.2. MI_DISP_LAYER

- Description

  Define the DISP_LAYER type.
  Video layer is the output unit of DISP. For DISP with multiple input ports, all input ports are eventually spliced into a video layer and output to the monitor or LCD. For DISP with only one input port, the input port size is equal to the video layer size.

- Definition

  typedef MI_S32  MI_DISP_LAYER;

- Member

  N/A.

- Related Data Type and Interface

  N/A.

## 3.3. MI_DISP_INPUTPORT

- Description

  Define the LAYER_INPUTPORT type.

- Definition

  typedef MI_S32 MI_DISP_INPUTPORT;

- Note

  N/A.

- Related Data Type and Interface

  N/A.

## 3.4. MI_DISP_Interface_e

- Description

  Define video output interface.

- Definition

```
typedef enum
{
    E_MI_DISP_INTF_CVBS = 0,
    E_MI_DISP_INTF_YPBPR,
    E_MI_DISP_INTF_VGA,
    E_MI_DISP_INTF_BT656,
    E_MI_DISP_INTF_BT1120,
    E_MI_DISP_INTF_HDMI,
    E_MI_DISP_INTF_LCD,
    E_MI_DISP_INTF_BT656_H,
    E_MI_DISP_INTF_BT656_L,
    E_MI_DISP_INTF_TTL,
    E_MI_DISP_INTF_MIPIDSI,
    E_MI_DISP_INTF_TTL_SPI_IF,
    E_MI_DISP_INTF_SRGB,
    E_MI_DISP_INTF_MCU,
    E_MI_DISP_INTF_MCU_NOFLM,
    E_MI_DISP_INTF_BT601,
    E_MI_DISP_INTF_BT1120_DDR,
    E_MI_DISP_INTF_MAX,
}MI_DISP_Interface_e;
```

- Note

  E_MI_DISP_INTF_BT1120 is single edge sampling; E_MI_DISP_INTF_BT1120_DDR is double edge sampling

- Related Data Type and Interface

MI_DISP_PubAttr_t
MI_DISP_SetPubAttr

## 3.5. MI_DISP_IntfSync_e

- Description

  define DISP timing type.

- Definition

```
typedef enum
{
    E_MI_DISP_OUTPUT_PAL = 0,
    E_MI_DISP_OUTPUT_NTSC,
    E_MI_DISP_OUTPUT_960H_PAL,  /* ITU-R BT.1302 960 x 576 at 50 Hz (interlaced)*/
    E_MI_DISP_OUTPUT_960H_NTSC, /* ITU-R BT.1302 960 x 480 at 60 Hz (interlaced)*/

    E_MI_DISP_OUTPUT_480i60,
    E_MI_DISP_OUTPUT_576i50,
    E_MI_DISP_OUTPUT_480P60,
    E_MI_DISP_OUTPUT_576P50,
    E_MI_DISP_OUTPUT_720P50,
    E_MI_DISP_OUTPUT_720P60,
    E_MI_DISP_OUTPUT_1080P24,
    E_MI_DISP_OUTPUT_1080P25,
    E_MI_DISP_OUTPUT_1080P30,
    E_MI_DISP_OUTPUT_1080I50,
    E_MI_DISP_OUTPUT_1080I60,
    E_MI_DISP_OUTPUT_1080P50,
    E_MI_DISP_OUTPUT_1080P60,

    E_MI_DISP_OUTPUT_640x480_60,   /* VESA 640 x 480 at 60 Hz (non-interlaced) CVT */
    E_MI_DISP_OUTPUT_800x600_60,   /* VESA 800 x 600 at 60 Hz (non-interlaced) */
    E_MI_DISP_OUTPUT_1024x768_60,  /* VESA 1024 x 768 at 60 Hz (non-interlaced) */
    E_MI_DISP_OUTPUT_1280x1024_60, /* VESA 1280 x 1024 at 60 Hz (non-interlaced) */
    E_MI_DISP_OUTPUT_1366x768_60,  /* VESA 1366 x 768 at 60 Hz (non-interlaced) */
    E_MI_DISP_OUTPUT_1440x900_60,  /* VESA 1440 x 900 at 60 Hz (non-interlaced) CVT
Compliant */
    E_MI_DISP_OUTPUT_1280x800_60,  /* 1280*800@60Hz VGA@60Hz*/
    E_MI_DISP_OUTPUT_1680x1050_60, /* VESA 1680 x 1050 at 60 Hz (non-interlaced) */
    E_MI_DISP_OUTPUT_1920x2160_30, /* 1920x2160_30 */
    E_MI_DISP_OUTPUT_1600x1200_60, /* VESA 1600 x 1200 at 60 Hz (non-interlaced) */
    E_MI_DISP_OUTPUT_1920x1200_60, /* VESA 1920 x 1600 at 60 Hz (non-interlaced) CVT
(Reduced Blanking)*/

    E_MI_DISP_OUTPUT_2560x1440_30,   /* 2560x1440_30 */
    E_MI_DISP_OUTPUT_2560x1440_50,   /* 2560x1440_50 */
    E_MI_DISP_OUTPUT_2560x1440_60,   /* 2560x1440_60 */
    E_MI_DISP_OUTPUT_2560x1600_60,   /* 2560x1600_60 */
    E_MI_DISP_OUTPUT_3840x2160_25,   /* 3840x2160_25 */
    E_MI_DISP_OUTPUT_3840x2160_30,   /* 3840x2160_30 */
    E_MI_DISP_OUTPUT_3840x2160_60,   /* 3840x2160_60 */
    E_MI_DISP_OUTPUT_1920x1080_5994, /* 1920x1080_59.94 */
```

```
E_MI_DISP_OUTPUT_1920x1080_2997, /* 1920x1080_29.97 */
E_MI_DISP_OUTPUT_1280x720_5994,  /* 1280X720_59.94 */
E_MI_DISP_OUTPUT_1280x720_2997,  /* 1280x720_29.97 */
E_MI_DISP_OUTPUT_3840x2160_2997, /* 3840x2160_29.97 */
E_MI_DISP_OUTPUT_720P24,         /* 1280x720_24 */
E_MI_DISP_OUTPUT_720P25,         /* 1280x720_25 */
E_MI_DISP_OUTPUT_720P30,         /* 1280x720_30 */
E_MI_DISP_OUTPUT_1920x1080_2398, /* 1920x1080_23.98*/
E_MI_DISP_OUTPUt_3840x2160_24,   /* 3840x2160_24 */
E_MI_DISP_OUTPUT_848x480_60,     /* 848x480_60 */
E_MI_DISP_OUTPUT_1280x768_60,    /* 1280x768_60 */
E_MI_DISP_OUTPUT_1280x960_60,    /* 1280x960_60 */
E_MI_DISP_OUTPUT_1360x768_60,    /* 1360x768_60 */
E_MI_DISP_OUTPUT_1400x1050_60,   /* 1400x1050_60 */
E_MI_DISP_OUTPUT_1600x900_60,    /* 1600x900_60 */
E_MI_DISP_OUTPUT_1920x1440_60,    /* 1920x1440_60 */

E_MI_DISP_OUTPUT_USER,
E_MI_DISP_OUTPUT_MAX,
 } MI_DISP_OutputTiming_e;
```

- Member

  For each Timing definition, LCD display should use E_MI_DISP_OUTPUT_USER.

- Note

- Related Data Type and Interface

  MI_DISP_PubAttr_t
  MI_DISP_SetPubAttr。

## 3.6. MI_DISP_SyncInfo_t

- Description

  Define the DISP sync information

- Definition

```
#define MI_VPE_MAX_PORT_NUM  (256)
typedef struct MI_DISP_SyncInfo_s
{
    MI_BOOL  bSynm;     /* sync mode (0:timing,as BT.656; 1:signal,as LCD) */
    MI_BOOL  bIop;      /* interlaced or progressive display (0:i; 1:p) */
    MI_U8    u8Intfb;   /* interlace bit width while output */

    MI_U16   u16VStart ;   /* vertical de start */
     MI_U16   u16Vact ;  /* vertical active area */
    MI_U16   u16Vbb;    /* vertical back blank porch */
    MI_U16   u16Vfb;    /* vertical front blank porch */

     MI_U16   u16HStart;   /* herizontal de start */
    MI_U16   u16Hact;   /* herizontal active area */
    MI_U16   u16Hbb;    /* herizontal back blank porch */
```

```
    MI_U16   u16Hfb;    /* herizontal front blank porch */
    MI_U16   u16Hmid;   /* bottom herizontal active area */

    MI_U16   u16Bvact;  /* bottom vertical active area */
    MI_U16   u16Bvbb;   /* bottom vertical back blank porch */
    MI_U16   u16Bvfb;   /* bottom vertical front blank porch */

    MI_U16   u16Hpw;    /* horizontal pulse width */
    MI_U16   u16Vpw;    /* vertical pulse width */

    MI_BOOL  bIdv;      /* inverse data valid of output */
    MI_BOOL  bIhs;      /* inverse horizontal synch signal */
    MI_BOOL  bIvs;      /* inverse vertical synch signal */
    MI_U32   u32FrameRate;
} MI_DISP_SyncInfo_t
```

- Member

| Member Name | Description |
|---|---|
| u16VStart | Start from the vertical effective. Generally equal to Vsync width + Vsync back porch |
| u16Vact | Line number of each frame |
| u16Vbb | Vsync back porch |
| u16Vfb | Vsync front porch |
| u16HStart | Start from the horizontal effective. Generally equal to Hsync width + Hsync back porch |
| u16Hact | Pixel count of each line |
| u16Hbb | Hsync back porch |
| u16Hfb | Hsync front porch |
| u16Hpw | Hsync width |
| u16Vpw | Vsync width |
| u32Framerate | Frame rate |

Parameters not listed in the table are parameters not used yet.

- Note

Only used when pictures are shown on LCD.

- Related Data Type and Interface

MI_DISP_PubAttr_t
MI_DISP_SetPubAttr

## 3.7. MI_DISP_PubAttr_t

- Description

Define the DISP public attribute

- Definition

typedef struct MI_DISP_PubAttr_s

```
{
    MI_U32     u32BgColor;   /* Background color of a device*/
    MI_DISP_Interface_e    eIntfType;        /* Type of a VO interface */
    MI_DISP_IntfSync_e     eIntfSync;        /* Type of a VO interface timing */
    MI_DISP_SyncInfo_t     stSyncInfo;       /* Information about VO interface timings */
} MI_DISP_PubAttr_t;
```

- Member

| Member | Description |
|---|---|
| u32BgColor | Set the background color |
| u32IntfType | Interface definition:<br>MI_DISP_Interface_e |
| eIntfSync | Interface timing definition:<br>MI_DISP_IntfSync_e |
| stSyncInfo | User-defined interface timing definition:<br>MI_DISP_SyncInfo_t |

- Note
  - stSyncInfo is used for E_MI_DISP_OUTPUT_USER; that is, the case in which the interface timing is user-defined.
  - Maruko u32BgColor format is xBGR, where x/B/G/R each occupies 8bit, and x is invalid data; The u32BgColor format of other chips is YUYV, in which Y/U/V each occupies 8 bits.

- Related Data Type and Interface

  MI_DISP_PubAttr_t
  MI_DISP_SetPubAttr

## 3.8. MI_DISP_DevAttachMode_e

- Description

  Define DISP video destination output mode.

- Definition

```
typedef enum
{
    E_MI_DISP_ATTACH_DEFAULT = 0,
    E_MI_DISP_ATTACH_OSD_MODE,
    E_MI_DISP_ATTACH_VIDEO_MODE,
}MI_DISP_DevAttachMode_e;
```

- Member

| Member | Description |
|---|---|
| E_MI_DISP_ATTACH_DEFAULT | Destination output default mode |
| E_MI_DISP_ATTACH_OSD_MODE | Output video screen with OSD superimposed |

| Member | Description |
|---|---|
| E_MI_DISP_ATTACH_VIDEO_MODE | Output video screen only |

- Note

  N/A.

- Related Data Type and Interface

  MI_DISP_DeviceAttach
  MI_DISP_DevAttachAttr_t

## 3.9. MI_DISP_DevAttachAttr_t

- Description

  Define DISP video destination output attribute.

- Definition

```
typedef struct MI_DISP_DevAttachAttr_s
{
    MI_DISP_DevAttachMode_e eDevAttachMode;
}MI_DISP_DevAttachAttr_t;
```

- Member

| Member | Description |
|---|---|
| eDevAttachMode | Set video destination output mode |

- Note

  N/A.

- Related Data Type and Interface

  MI_DISP_DeviceAttach

## 3.10. MI_DISP_Csc_t

- Description

  Define the DISP color space conversion information

- Definition

```
typedef struct MI_DISP_Csc_s
{
    MI_DISP_CscMattrix_e eCscMatrix;
    MI_U32 u32Luma;              /* luminance:   0 ~ 100 default: 50 */
    MI_U32 u32Contrast;          /* contrast :   0 ~ 100 default: 50 */
    MI_U32 u32Hue;               /* hue      :   0 ~ 100 default: 50 */
    MI_U32 u32Saturation;        /* saturation:  0 ~ 100 default: 50 */
} MI_DISP_Csc_t;
```

- Member

| Member | Description |
|---|---|
| eCscMatrix | Color space conversion matrix |
| u32Luma | Luminance adjustment |
| u32Contrast | Contrast adjustment |
| u32Hue | Hue adjustment |
| u32Saturation | Saturation adjustment |

- Note

  Currently supported chips include MSR930, MSR650x, Taiyaki, and Takoyaki series.

- Related Data Type and Interface

  MI_DISP_GetVgaParam
  MI_DISP_GetVgaParam
  MI_DISP_GetHdmiParam
  MI_DISP_SetHdmiParam
  MI_DISP_GetLcdParam
  MI_DISP_SetLcdParam

## 3.11. MI_DISP_CscMattrix_e

- Description

  Define the DISP color space conversion matrix

- Definition

```
typedef enum
{
    E_MI_DISP_CSC_MATRIX_BYPASS = 0,
     /* do not change color space */
    E_MI_DISP_CSC_MATRIX_BT601_TO_RGB_PC,
      /* change color space from BT.601 to RGB */
    E_MI_DISP_CSC_MATRIX_BT709_TO_RGB_PC,
      /* change color space from BT.709 to RGB */
    E_MI_DISP_CSC_MATRIX_RGB_TO_BT601_PC,
     /* change color space from RGB to BT.601 */
    E_MI_DISP_CSC_MATRIX_RGB_TO_BT709_PC,
        /* change color space from RGB to BT.709 */
     E_MI_DISP_CSC_MATRIX_NUM
} MI_DISP_CscMatrix_e;
```

- Member

| Member | Description |
|---|---|
| E_MI_DISP_CSC_MATRIX_BYPASS | Bypass color space conversion |
| E_MI_DISP_CSC_MATRIX_BT601_TO_RGB_PC | BT.601 to RGB color space CSC matrix |

| | |
|---|---|
| E_MI_DISP_CSC_MATRIX_BT709_TO_RGB_PC | BT.709 to RGB color space CSC matrix |
| E_MI_DISP_CSC_MATRIX_RGB_TO_BT601_PC | RGB to BT.601 color space CSC matrix |
| E_MI_DISP_CSC_MATRIX_RGB_TO_BT709_PC | RGB to BT.709 color space CSC matrix |

- Note

  Currently supported chips include MSR930, MSR650x, Taiyaki, and Takoyaki series.

- Related Data Type and Interface

  MI_DISP_GetVgaParam
  MI_DISP_GetVgaParam
  MI_DISP_GetHdmiParam
  MI_DISP_SetHdmiParam
  MI_DISP_GetLcdParam
  MI_DISP_SetLcdParam

## 3.12. MI_DISP_VgaParam_t

- Description

  Define the DISP VGA parameter information

- Definition

```
typedef struct MI_DISP_VgaParam_s
{
    MI_DISP_Csc_t stCsc;              /* color space */
    MI_U32 u32Gain;                   /* current gain of VGA signals. [0, 64). default:0x30
 */
    MI_U32 u32Sharpness;
} MI_DISP_VgaParam_t;
```

- Member

| Member | Description |
|---|---|
| stCsc | Color space conversion information definition |
| u32Gain | Signal strength gain |
| u32Sharpness | Sharpness definition |

- Note

  Currently supported chips include MSR930, MSR650x, Taiyaki series.

- Related Data Type and Interface

  MI_DISP_Csc_t
  MI_DISP_GetVgaParam

MI_DISP_GetVgaParam

## 3.13. MI_DISP_HdmiParam_t

- Description

  Define the DISP HDMI parameter information

- Define

```
typedef struct MI_DISP_HdmiParam_s
{
MI_DISP_Csc_t stCsc;                /* color space */
MI_U32 u32Gain;         /* current gain of HDMI signals. [0, 64). default:0x30 */
MI_U32 u32Sharpness;
} MI_DISP_HdmiParam_t;
```

- Member

| Member | Description |
|---|---|
| stCsc | Color space conversion information definition |
| u32Gain | Signal strength gain |
| u32Sharpness | Definition of sharpness |

- Note

  Currently supported chips include MSR930, MSR650x,  Taiyaki series.

- Related Data Type and Interface

  MI_DISP_Csc_t
  MI_DISP_GetHdmiParam
  MI_DISP_SetHdmiParam

## 3.14. MI_DISP_LcdParam_t

- Description

  Define the DISP LCD parameter information

- Definition

```
typedef struct MI_DISP_LcdParam_s
{
    MI_DISP_Csc_t stCsc;                /* color space */
    MI_U32 u32Sharpness;
} MI_DISP_LcdParam_t;
```

- Member

| Member | Description |
|---|---|

| | |
|---|---|
| stCsc | Color space conversion information definition |
| u32Sharpness | Sharpness definition |

- Note

  Currently supported chips include Takoyaki series.

- Related Data Type and Interface

  MI_DISP_Csc_t
  MI_DISP_GetLcdParam
  MI_DISP_SetLcdParam

## 3.15. MI_DISP_CvbsParam_t

- Description

  Define the DISP CVBS output image parameter

- Definition

```
typedef struct MI_DISP_CvbsParam_s
{
    MI_DISP_Csc_t stCsc;                /* color space */
    MI_U32 u32Sharpness;
} MI_DISP_CvbsParam_t;
```

- Member

| Member | Description |
|---|---|
| stCsc | Color conversion information definition |
| u32Sharpness | Sharpness definition |

- Note

  N/A.

- Related Data Type and Interface

  MI_DISP_Csc_t
  MI_DISP_GetCvbsParam
  MI_DISP_SetCvbsParam

## 3.16. MI_DISP_ColorTemperature_t

- Description

  Define the DISP color temperature parameter

- Definition

```
typedef struct
```

```
{
    MI_U16 u16RedOffset;
    MI_U16 u16GreenOffset;
    MI_U16 u16BlueOffset;

    MI_U16 u16RedColor;  // 00~FF, 0x80 is no change
    MI_U16 u16GreenColor;// 00~FF, 0x80 is no change
    MI_U16 u16BlueColor; // 00~FF, 0x80 is no change
}MI_DISP_ColorTemperature_t;
```

- Member

| Member | Description |
|---|---|
| u16RedColor | Red component definition |
| u16GreenColor | Green component definition |
| u16BlueColor | Blue component definition |

- Note

  Currently supported chips include Takoyaki series.

- Related Data Type and Interface

  MI_DISP_DeviceGetColorTempeture
  MI_DISP_DeviceSetColorTempeture

## 3.17. MI_DISP_GammaParam_t

- Description

  Define the DISP Gamma parameter

- Definition

```
typedef struct
{
    MI_BOOL bEn;
    MI_U16 u16EntryNum;
    MI_U8 *pu8ColorR;
    MI_U8 *pu8ColorG;
    MI_U8 *pu8ColorB;
}MI_DISP_GammaParam_t;
```

- Member

| Member | Description |
|---|---|
| bEn | Enable gamma adjustment |
| u16EntryNum | The number of members in gamma table |
| pu8ColorR | Pointer to red component gamma table |

| | |
|---|---|
| pu8ColorG | Pointer to green component gamma table |
| pu8ColorB | Pointer to blue component gamma table |

- Note

  Currently supported chips include Takoyaki series.

- Related Data Type and Interface

  MI_DISP_DeviceSetGammaParam

## 3.18. MI_DISP_VideoLayerSize_t

- Description

  Define the DISP video layer size

- Definition

```
typedef struct MI_DISP_VideoLayerSize_s
{
    MI_U32 u32Width;
    MI_U32 u32Height;
} MI_DISP_VideoLayerSize_t;
```

- Member

| Member | Description |
|---|---|
| u32Width | X-axis width |
| u32Height | Y-axis height |

- Note

  N/A.

- Related Data Type and Interface

  MI_DISP_SetVideoLayerAttr
  MI_DISP_GetVideoLayerAttr

## 3.19. MI_DISP_VideoLayerAttr_t

- Description

  Define the DISP video layer attribute

- Definition

```
typedef struct MI_DISP_VideoLayerAttr_s
{
    MI_DISP_VidWinRect_t     stDispWin;       /* Display resolution */
    MI_DISP_VideoLayerSize_t  stLayerSize;     /* Canvas size of the video layer */
    MI_DISP_PixelFormat_e  ePixFormat;       /* Pixel format of the video layer */
} MI_DISP_VideoLayerAttr_t;
```

- Member

| Member | Description |
|---|---|
| stDispWin | Input port display position |
| stLayerSize | Video layer size |
| ePixFormat | Input pixel data format |

- Note

N/A.

- Related Data Type and Interface

MI_DISP_SetVideoLayerAttr
MI_DISP_GetVideoLayerAttr

## 3.20. MI_DISP_RotateMode_e

- Description

Define the DISP rotation angle information

- Definition

```
typedef enum
{
    E_MI_DISP_ROTATE_NONE,
      E_MI_DISP_ROTATE_90,
      E_MI_DISP_ROTATE_180,
    E_MI_DISP_ROTATE_270,
    E_MI_DISP_ROTATE_NUM,
}MI_DISP_RotateMode_e;
```

- Member

| Member | Description |
|---|---|
| E_MI_DISP_ROTATE_NONE | Do not rotate |
| E_MI_DISP_ROTATE_90 | Rotate 90 degrees clockwise |
| E_MI_DISP_ROTATE_180 | Rotate 180 degrees clockwise |
| E_MI_DISP_ROTATE_270 | Rotate 270 degrees clockwise |

- Note

  - Currently supported chip series include Takoyaki/ Tiramisu/ Muffin.
  - Only Takoyaki supports rotation by 180 degrees.

- Related Data Type and Interface

MI_DISP_SetVideoLayerRotateMode

## 3.21. MI_DISP_RotateConfig_t

- Description

  Define the DISP rotation parameter

- Definition

```
typedef struct MI_DISP_RotateConfig_s
{
    MI_DISP_RotateMode_e eRotateMode;
}MI_DISP_RotateConfig_t;
```

- Member

| Member | Description |
|---|---|
| eRotateMode | rotation angle enum |

- Note
  - Currently supported chip series include Takoyaki/ Tiramisu/ Muffin.
  - Only Takoyaki supports rotation by 180 degrees.

- Related Data Type and Interface

  MI_DISP_SetVideoLayerRotateMode

## 3.22. MI_DISP_VidWinRect_t

- Description

  Define the DISP window attribute

- Definition

```
typedef struct MI_DISP_VidWin_Rect_s
{
    MI_U16 u16X;
    MI_U16 u16Y;
    MI_U16 u16Width;
    MI_U16 u16Height;
} MI_DISP_VidWinRect_t;
```

- Member

| Member | Description |
|---|---|
| u16X | X-axis start point |
| u16Y | Y-axis start point |
| u16Width | X-axis width |
| u16Height | Y-axis height |

- Note
  - x, y, width and height of each window must be aligned with 2pixels.
  - Cropping cannot exceed the source area.

- Related Data Type and Interface
  MI_DISP_SetZoomInWindow

## 3.23. MI_DISP_InputPortAttr_t

- Description
  Define the DISP input port attribute

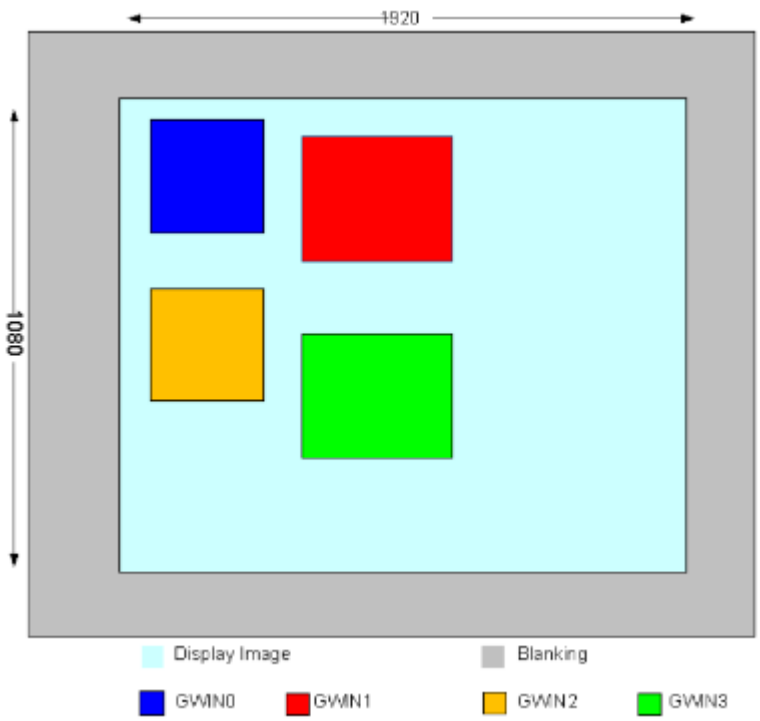
- Definition

```
typedef struct MI_DISP_InputPortAttr_s
{
    MI_DISP_VidWinRect_t stDispWin;                 /* rect of video out chn */
    MI_U16 u16SrcWidth;
      MI_U16 u16SrcHeight;
      MI_SYS_CompressMode_e eDecompressMode;
} MI_DISP_InputPortAttr_t;
```

- Member

| Member | Description |
|---|---|
| stDispWin | Specify input port output window position and size |
| u16SrcWidth | Specify input image width |
| u16SrcHeight | Specify input image height |
| eDecompressMode | Specify input decompression mode |

- Note
  - The Takoyaki/ Taiyaki/ Tiramisu/ Muffin/ Mochi series chips can can scale up for each port (scaling down is not supported), and the maximum magnification in the H/V direction is 16 times.
  - If the chip cannot scale up the port independently, u16SrcWidth should be equal to u16Width defined in stDispWin, and u16SrcHeight should be equal to u16Height defined in stDispWin.
  - For a video layer with multiple input ports, the display position of each port on the video layer can be set arbitrarily.
  - The display position of the input port cannot be superimposed with the enabled port (including Hide), but the disabled port can be superimposed
  - x, y, width and height of each window must be aligned with 2pixels.
  - About eDecompressMode, Mochi series chips support E_MI_SYS_COMPRESS_MODE_NONE/E_MI_SYS_COMPRESS_MODE_TO_6BIT, other chips only support E_MI_SYS_COMPRESS_MODE_NONE.




- Related Data Type and Interface

  MI_DISP_SetInputPortAttr
  MI_DISP_GetInputPortAttr

## 3.24. MI_DISP_Position_t

- Description

  Define the DISP input port position

- Definition

```
typedef struct MI_DISP_Position_s
{
    MI_U16 u16X;
    MI_U16 u16Y;
} MI_DISP_Position_t;
```

- Member

| Member | Description |
|---|---|
| u16X | X-axis start point |
| u16Y | Y-axis start point |

- Note

  N/A.

- Related Data Type and Interface

MI_DISP_GetInputPortDispPos
MI_DISP_SetInputPortDispPos

## 3.25. MI_DISP_SyncMode_e

- Description

  Define the DISP sync mode

- Definition

```
typedef enum
{
    E_MI_DISP_SYNC_MODE_INVALID = 0,
    E_MI_DISP_SYNC_MODE_CHECK_PTS,
    E_MI_DISP_SYNC_MODE_FREE_RUN,
    E_MI_DISP_SYNC_MODE_NUM,
} MI_DISP_SyncMode_e;
```

- Member

| Member | Description |
|---|---|
| E_MI_DISP_SYNC_MODE_INVALID | Invalid sync mode |
| E_MI_DISP_SYNC_MODE_CHECK_PTS | Check the PTS sync mode |
| E_MI_DISP_SYNC_MODE_FREE_RUN | Bypass the PTS sync mode |

- Note

  Currently supported chips include MSR930 and MSR650x.

- Related Data Type and Interface

  MI_DISP_SetInputPortSyncMode

## 3.26. MI_DISP_InputPortStatus_e

- Description

  Define the DISP input port status type

- Definition

```
typedef enum
{
    E_MI_DISP_INPUTPORT_STATUS_INVALID = 0,
    E_MI_DISP_INPUTPORT_STATUS_PAUSE,
    E_MI_DISP_INPUTPORT_STATUS_RESUME,
    E_MI_DISP_INPUTPORT_STATUS_STEP,
    E_MI_DISP_INPUTPORT_STATUS_REFRESH,
    E_MI_DISP_INPUTPORT_STATUS_SHOW,
    E_MI_DISP_INPUTPORT_STATUS_HIDE,
    E_MI_DISP_ INPUTPORT_STATUS _NUM,
} MI_DISP_InputPortStatus_e;
```

- Member

| Member | Description |
|---|---|
| E_MI_DISP_INPUTPORT_STATUS_INVALID | Invalid status mode |
| E_MI_DISP_INPUTPORT_STATUS_PAUSE | Pause status |
| E_MI_DISP_INPUTPORT_STATUS_RESUME | Resume status |
| E_MI_DISP_INPUTPORT_STATUS_STEP | Step status |
| E_MI_DISP_INPUTPORT_STATUS_REFRESH | Refresh status |
| E_MI_DISP_INPUTPORT_STATUS_SHOW | Show status |
| E_MI_DISP_INPUTPORT_STATUS_HIDE | Hide status |

- Note

  N/A.

- Related Data Type and Interface

  N/A.

## 3.27. MI_DISP_QueryChannelStatus_t

- Description

  Define the DISP input port status information

- Definition

```
typedef struct MI_DISP_QueryChannelStatus_s
{
    MI_BOOL bEnable;
    MI_DISP_InputPortStatus_e eStatus;
} MI_DISP_QueryChannelStatus_t;
```

- Member

| Member | Description |
|---|---|
| bEnable | Input port enable status |
| eStatus | Input port status type |

- Note

  N/A.

- Related Data Type and Interface

  MI_DISP_InputPortStatus_e
  MI_DISP_QueryInputPortStat

## 3.28. MI_DISP_VideoFrame_t

- Description

Define the DISP video frame information.

- Definition

```
typedef struct MI_DISP_VideoFrame_s
{
    MI_U32          u32Width;
    MI_U32          u32Height;
    MI_DISP_PixelFormat_e  ePixelFormat;
    MI_PHY          aphyAddr[3];
    Void *          *pavirAddr[3];
    MI_U64          u64pts;
    MI_U32          u32PrivateData;
} MI_DISP_VideoFrame_t;
```

- Member

| Member | Description |
|---|---|
| u32Width | Image width |
| u32Height | Image height |
| ePixelFormat | Pixel format |
| u32PhyAddr | Image data physical address |
| pVirAddr | Image data virtual address |
| u64pts | Image data timestamp |
| u32PrivateData | Private data |

- Note

Currently supported chips include MSR930 and MSR650x.

- Related Data Type and Interface

MI_DISP_GetScreenFrame
MI_DISP_ReleaseScreenFrame

## 3.29. MI_DISP_InitParam_t

- Description

DISP device initialization parameter.

- Definition

```
typedef struct MI_DISP_InitParam_s
{
    MI_U32 u32DevId;
    MI_U8 *u8Data;
} MI_DISP_InitParam_t;
```

- Member

| Member | Description |
|---|---|
| u32DevId | Device ID |

| | |
|---|---|
| u8Data | Data pointer buffer |

- Note

  N/A.

- Related Data Type and Interface

  MI_DISP_InitDev

## 3.30. MI_DISP_WBC

- Description

  Define the type of DISP_WBC.

- Definition

  typedef MI_S32 MI_DISP_WBC

## 3.31. MI_DISP_WBC_SourceType_e

- Description

  The source type of the data captured by the video write-back device, used to distinguish whether it is from the video layer or the device.

- Definition

```
typedef enum {
    MI_DISP_WBC_SOURCE_DEV,
    MI_DISP_WBC_SOURCE_VIDEO,
}MI_DISP_WBC_SourceType_e;
```

- Member

| Member | Description |
|---|---|
| MI_DISP_WBC_SOURCE_DEV | The write back source is device (video + UI) |
| MI_DISP_WBC_SOURCE_VIDEO | The write back source is video layer (video only) |

- Note

  N/A.

- Related Data Type and Interface

  MI_DISP_SetWBCSource
  MI_DISP_GetWBCSource

## 3.32. MI_DISP_WBC_Source_t

- Description

  The write-back source of video write-back device.

- Definition

```
typedef struct MI_DISP_WBC_Source_s
{
    MI_DISP_WBC_SourceType_e eSourceType;
    MI_U32 u32SourceId;
}MI_DISP_WBC_Source_t;
```

- Member

| Member | Description |
|---|---|
| eSourceType | Write back source type |
| u32SourceId | Write back source device ID |

- Note

N/A.

- Related Data Type and Interface

MI_DISP_SetWBCSource
MI_DISP_GetWBCSource

## 3.33. MI_DISP_WBC_TargetSize_t

- Description

The target size of the video write-back device.

- Definition

```
typedef struct MI_DISP_WBC_TargetSize_s
{
    MI_U32 u32Width;
    MI_U32 u32Height;
}MI_DISP_WBC_TargetSize_t;
```

- Member

| Member | Description |
|---|---|
| u32Width | Width of write-back target |
| u32Height | Height of write-back target |

- Note

WBC supports zooming out, but not zooming in. If the target resolution is smaller than the source resolution, it will be automatically zooming out.

- Related Data Type and Interface

MI_DISP_SetWBCAttr
MI_DISP_GetWBCAttr

## 3.34. MI_DISP_WBC_Attr_t

- Description

  Video write-back device attributes.

- Definition

```
typedef struct MI_DISP_WBC_Attr_s
{
    MI_DISP_WBC_TargetSize_t stTargetSize;
      MI_SYS_PixelFormat_e ePixFormat;
      MI_SYS_CompressMode_e eCompressMode;
}MI_DISP_WBC_Attr_t;
```

- Member

| Member | Description |
|---|---|
| stTargetSize | Write-back target size |
| ePixFormat | Pixel format of write-back image |
| eCompressMode | Compress mode |

- Note

  eDecompressMode is only supported by Mochi series chips.

- Related Data Type and Interface

  MI_DISP_SetWBCAttr

  MI_DISP_GetWBCAttr

# 4. DISP ERROR CODE

The following table lists the DISP API error codes.

Table 4-3: DISP API Error Codes

| Error Code | Macro Definition | Description |
|---|---|---|
| 0xA00F2001 | MI_ERR_DISP_INVALID_DEVID | Invalid Device ID |
| 0xA00F2002 | MI_ERR_DISP_INVALID_CHNID | Invalid Input Port ID |
| 0xA00F2003 | MI_ERR_DISP_ILLEGAL_PARAM | Illegal parameter |
| 0xA00F2006 | MI_ERR_DISP_NULL_PTR | Null pointer in function parameter |
| 0xA00F2008 | MI_ERR_DISP_NOT_SUPPORT | Operation not supported |
| 0xA00F2009 | MI_ERR_DISP_NOT_PERMIT | Operation not permitted |
| 0xA00F200C | MI_ERR_DISP_NO_MEM | No memory |
| 0xA00F2010 | MI_ERR_DISP_SYS_NOTREADY | System not ready |
| 0xA00F2012 | MI_ERR_DISP_BUSY | System busy |
| 0xA00F2040 | MI_ERR_DISP_DEV_NOT_CONFIG | Device not configured |
| 0xA00F2041 | MI_ERR_DISP_DEV_NOT_ENABLE | Device not enabled |
| 0xA00F2042 | MI_ERR_DISP_DEV_ALREADY_ENABLED | Device has been enabled already |
| 0xA00F2043 | MI_ERR_DISP_DEV_ALREADY_BOUND | Device has been bound already |
| 0xA00F2044 | MI_ERR_DISP_DEV_NOT_BIND | Device not bound |
| 0xA00F2045 | MI_ERR_DISP_LAYER_NOT_ENABLE | Video layer not enabled |
| 0xA00F2046 | MI_ERR_DISP_LAYER_NOT_DISABLE | Video layer not disabled |
| 0xA00F2047 | MI_ERR_DISP_LAYER_NOT_CONFIG | Video layer not configured |
| 0xA00F2048 | MI_ERR_DISP_INPUTPORT_NOT_DISABLE | Input port not disabled |
| 0xA00F2049 | MI_ERR_DISP_INPUTPORT _NOT_ENABLE | Input port not enabled |
| 0xA00F204A | MI_ERR_DISP_INPUTPORT _NOT_CONFIG | Input port not configured |
| 0xA00F204B | MI_ERR_DISP_INPUTPORT _NOT_ALLOC | Input port not allocated |
| 0xA00F204C | MI_ERR_DISP_INVALID_PATTERN | Invalid pattern |
| 0xA00F204D | MI_ERR_DISP_INVALID_POSITION | Invalid position |
| 0xA00F204E | MI_ERR_DISP_WAIT_TIMEOUT | Wait timeout |
| 0xA00F204F | MI_ERR_DISP_INVALID_VIDEO_FRAME | Invalid video frame |
| 0xA00F2050 | MI_ERR_DISP_INVALID_RECT_PARA | Invalid rectangle parameter |
| 0xA00F2051 | MI_ERR_DISP_ INPUTPORT_SHOW_AREA_OVERLAP | Input port area overlapped |

| Error Code | Macro Definition | Description |
|---|---|---|
| 0xA00F2052 | MI_ERR_DISP_INVALID_LAYERID | Invalid video layer ID |
| 0xA00F2053 | MI_ERR_DISP_LAYER_ALREADY_BOUND | Video layer has been bound already |
| 0xA00F2054 | MI_ERR_DISP_LAYER_NOT_BIND | Video layer not bound |

# 5. PROCFS INTRODUCTION

## 5.1. DISP cat

- Debug Information

  # cat /proc/mi_modules/mi_disp/mi_disp0

```
============================== disp dev0 info ==============================
 DevStatus         IrqNum         IrqCnt        BgColor
         1             52            700         800080
 Interface      DevTiming      CscMatrix           Luma       Contrast            Hue     Saturation      Sharpness           Gain
      HDMI        720P60              3             50             50             50             50              0              0
======================================== disp layer[0] Info ========================================
   LayerId     BindedDevID     LayerWidth    LayerHeight
         0               0              0              0
   LayerId   LayDispWidth  LayDispHeight     Toleration     rotatemode
         0            1280            720              0           NONE
    PortId     enable CurStatus      src_w      src_h     crop_x     crop_y     crop_w     crop_h     show_x     show_y     show_w     show_h
         0            1         0        200        120          0          0        200        120          0          0        640        720
         1            1         0        200        120          0          0        200        120        640          0        640        720
    PortId      RecvBufCnt      RecvBuf_W      RecvBuf_H      Content_W      Content_H  RecvBufStride         PixFmt       syncmode
         0             539            200            120            200            120            208 semiplanar_420        Invalid
         1             527            200            120            200            120            208 semiplanar_420        Invalid
    PortId    OnScreenTask      FiredTask    StepTaskCnt  bClearAllTask            fps
         0        c2c5f940         (null)              0              0             25
         1        c2c5f580        c2c5fac0             0              0             25
```

- Debug Information Analysis

  Record the current usage status of DISP, device attribute, layer attribute, and inputport attribute. This information can be obtained dynamically, which is convenient for debugging and testing.

- Parameter Description

| Parameter | | Description |
|---|---|---|
| device info | DevStatus | Enable or Disable<br>0: Disable<br>1: Enable |
| | IrqNum | Interrupt Number |
| | IrqCnt | Interrupt Count |
| | BgColor | Background Color (YUYV accounts for 8bit) |
| | Interface | Interface Type<br>CVBS, YPBPR, VGA, BT656, BT1120, MCU, MCU_NOFLM, HDMI, LCD, BT656_H, BT656_L, TTL, MIPI_DSI, TTL_SPI, SRGB |
| | DevTiming | Output Timing, range:<br>[E_MI_DISP_OUTPUT_PAL ~ E_MI_DISP_OUTPUT_USER] |
| | CscMatrix | CSC Matrix selection, range:<br>[E_MI_DISP_CSC_MATRIX_BYPASS ~ E_MI_DISP_CSC_MATRIX_RGB_TO_BT709_PC] |
| device info | Luma | [0 ~ 99], default 50 |
| | Contrast | [0 ~ 99], default 50 |
| | Hue | [0 ~ 99], default 50 |
| | Saturation | [0 ~ 99], default 50 |
| | Sharpness | [0 ~ 255], default 50 |
| | Gain | [0 ~ 255], default 50 |

| Parameter | | Description |
|---|---|---|
| layer info | LayerId | The value range depends on the chip specifications. |
| | BindedDevID | The value range depends on the chip specifications. |
| | LayerWidth | Layer Width |
| | LayerHeight | Layer Height |
| | LayDispWidth | Layer Display Width |
| | LayDispHeight | Layer Display Height |
| | Toleration | PTS error tolerance threshold, in milliseconds |
| | rotatemode | Parameter range:<br>[E_MI_DISP_ROTATE_NONE~E_MI_DISP_ROTATE_270] |
| port info | PortId | The value range depends on the chip specifications. |
| | enable | Enable or Disable<br>0: Disable<br>1: Enable |
| | CurStatus | [invalid, pause, resume, step, refresh, show, hide]<br>The default is invalid. |
| | src_w | The width of original image |
| | src_h | The height of original image |
| | crop_x | The starting abscissa of crop area |
| | crop_y | The starting ordinate of crop area |
| | crop_w | The width of crop area |
| | crop_h | The height of crop area |
| | show_x | The starting abscissa of the port on the layer |
| | show_y | The starting ordinate of the port on the layer |
| | show_w | The width of the port display |
| | show_h | The height of the port display |
| | RecvBufCnt | The count of buffs currently received |
| | RecvBuf_W | The width of input image |
| | RecvBuf_H | The height of input image |
| | Content_W | The valid width of input image |
| | Content_H | The valid height of input image |
| | RecvBufStride | Stride of input image |
| | PixFmt | Pixel format of input image |
| | syncmode | Check PTS/Free Run<br>Parameter range: [Invalid, CheckPts, FreeRun] |
| | OnScreenTask | The Task currently being displayed, if it is NULL, it may be that no buff was sent to disp, or disp was not interrupted, or the cmdq of disp did not take effect. |
| | FiredTask | Task to be displayed |
| | StepTaskCnt | Number of Task to be processed step by step |
| | bClearAllTask | Whether to clear all Task<br>0: not clear<br>1: clear all Task |

| Parameter | | Description |
|---|---|---|
| | fps | Frame rate |

## 5.2. DISP echo

| | |
|---|---|
| Function | Get the echo command supported by the current DISP and its specific usage form. |
| Command | echo help > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | N/A |
| Example | echo help > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Get current screen data (reserved, currently not supported). |
| Command | echo getcapframe [devid] [layerid] [path] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [devid] Device id<br>[layerid] Video layer id<br>[path] Path and file name |
| Example | echo getcapframe 0 0 /mnt/frame.yuv > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Open/Close pts check of each port. |
| Command | echo checkframepts [layerid] [protid] [ON/OFF] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[portid] port id<br>[ON/OFF] ON: open, OFF: close |
| Example | echo checkframepts 0 0 ON > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Dump frame data. |
| Command | echo dumpframe [layerid] [portid] [path] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[portid] port id<br>[path] Storage path of dumped data |
| Example | echo dumpframe 0 0 /mnt/ > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Stop single port of disp to get buff. |
| Command | echo stopgetbuff [layerid] [portid] [ON/OFF] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[portid] port id<br>[ON/OFF] ON: stop, OFF: resume |
| Example | echo stopgetbuff 0 0 ON > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Set background color. |
| Command | echo bgcolor [devid] [value] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [devid] Device id<br>[value] color value |
| Example | echo setbgcolor 0 255 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Dynamically set screen effects. |
| Command | echo csc [devid] [CscMatrix] [Contrast] [Hue] [Luma] [Saturation] [Sharpness] [Gain] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [devid] Device id<br>[CscMatrix] Color matrix<br>[Contrast] Contrast<br>[Hue] Hue<br>[Luma] Luma<br>[Saturation] Saturation<br>[Sharpness] Sharpness<br>[Gain] Gain |
| Example | echo csc 0 0 50 50 50 50 0 0 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Debug color temperature. |
| Command | echo colortemp [devid] [BlueOffset] [GreenOffset] [RedOffset] [BlueColor] [GreenColor] [RedColor] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [devid] Device id<br>[Blue/Green/Redoffset] Not used<br>[RedColor] R Portion (0-255)<br>[GreenColor] G Portion (0-255)<br>[BlueColor]B Portion (0-255) |
| Example | echo setcolortemp 0 50 50 50 50 50 50 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Rotate the layer. |
| Command | echo rotate [layerid] [0/1/2]> /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[0/1/2] rotate configuration:<br>0: E_MI_DISP_ROTATE_NONE, not rotate<br>1: E_MI_DISP_ROTATE_90, rotate 90°<br>2: E_MI_DISP_ROTATE_270, rotate 270° |
| Example | echo rotate 1 1 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Crop the original data of a port. |
| Command | echo crop [layerid] [portid] [x] [y] [width] [height]> /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[portid] port id<br>[x] The starting abscissa of crop area<br>[y] The starting ordinate of crop area<br>[width] The width of crop area<br>[height] The height of crop area |
| Example | echo crop 0 0 0 0 100 100 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Hide a port. |
| Command | echo hide [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[portid] port id |
| Example | echo hide 0 0 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Show a port. |
| Command | echo show [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[portid] port id |
| Example | echo show  0 0 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Stop getting data for a port. |
| Command | echo pause [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[portid] port id |
| Example | echo pause 0 0 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Resume getting data for a port. |
| Command | echo resume [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[portid] port id |
| Example | echo resume 0 0 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Control a port to stop getting data after acquiring a frame of data. |
| Command | echo step [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter | [layerid] Video layer id |

| | |
|---|---|
| Description | [portid] port id |
| Example | echo step 0 0 > /proc/mi_modules/mi_disp/mi_disp0 |

| | |
|---|---|
| Function | Clear data of a port. |
| Command | echo clear [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0 |
| Parameter Description | [layerid] Video layer id<br>[portid] port id |
| Example | echo clear 0 0 > /proc/mi_modules/mi_disp/mi_disp0 |

## 5.3. WBC cat

- Debug Information

  # cat proc/mi_modules/mi_disp/mi_wbc0

```
==============================wbc0==============================
         irqnum         isrcnt     irq enable
             55           1198              1
        src dev           srcw           srch       src type        targetW        targetH
              1           1920           1080              1            200            120
    OnScreenTask      FiredTask            fps
        c3a57c38       c3a57438             20
/ #
```

- Debug Information Analysis

  Record the current use status and data attributes of WBC, and these information can be obtained dynamically to facilitate debugging and testing.

- Parameter Description

| Parameter | | Description |
|---|---|---|
| device info | irqnum | Interrupt Number |
| | Isrcnt | Interrupt Count |
| | irq enable | Interrupt Status:<br>0: disable<br>1: enable |
| | src dev | src device number |
| | srcw | The width of data source |
| | srch | The height of data source |
| | src type | Data source type<br>0: video+ui<br>1: video |
| | targetW | The width of data output |
| | targetH | The height of data output |
| | OnScreenTask | Task currently being output |

| Parameter | | Description |
|---|---|---|
| | FiredTask | Task to be output |

## 5.4. WBC echo

| | |
|---|---|
| Function | Get the echo command supported by the current WBC and its specific usage form. |
| Command | echo help > /proc/mi_modules/mi_disp/mi_wbc0 |
| Parameter Description | N/A |
| Example | echo help > /proc/mi_modules/mi_disp/mi_wbc0 |

| | |
|---|---|
| Function | dump frame data |
| Command | echo dumpframe [path] > /proc/mi_modules/mi_disp/mi_wbc0 |
| Parameter Description | [path] Storage path of dumped data |
| Example | echo dumpframe /mnt/ > /proc/mi_modules/mi_disp/mi_wbc0 |